Lévy processes

1. Introduction

The key properties that characterise Brownian motion are the independence and stationarity of its increments, together with the continuity of its paths. What happens if we drop the last requirement and allow the process to have jumps? We shall see that this leads to the class of Lévy processes.

Such processes are used, for example, in the modelling of

(1) insurance companies;
(2) markets subject to shocks and large investors;
(3) the motion of particles in an inhomogeneous medium;
(4) the movement of predators.

We briefly recall the notion of Poisson random measure and compound Poisson process.

2. Poisson random measures and compound Poisson processes

Let $\mathbb{U}$ be a given set, for example $\mathbb{U} = \mathbb{R}$, $\mathbb{U} = \mathbb{R} \times \mathbb{R}^+$, or $\mathbb{U} = \mathbb{R}^+ \times \{1\}$, and let $(\mathbb{U}, \mathcal{U}, \nu)$ be a measure space; that is, $\mathcal{U}$ is a sigma-algebra on $\mathbb{U}$, and ν is a sigma-finite measure on $\mathcal{U}$. Given a probability space $\{\Omega, \mathcal{F}, \mathbb{P}\}$, we say that $\mu : \Omega \times \mathcal{U} \to \mathbb{N} \cup \{\infty\}$ is a Poisson random measure if:

(1) for every fixed $A \in \mathcal{U}$, $\mu(\cdot, A)$ is a random variable on $\{\Omega, \mathcal{F}, \mathbb{P}\}$, and $\mathbb{P}$-almost surely $\mu(\omega, \cdot)$ is a measure; that is, μ is a random measure;
(2) for every $n \geq 2$ and every collection of disjoint sets $A_1, \ldots, A_n \in \mathcal{U}$, the random variables $(\mu(\cdot, A_1), \ldots, \mu(\cdot, A_n))$ are mutually independent;
(3) for every $A \in \mathcal{U}$, $\mu(\cdot, A) \sim Poi(\nu(A))$, with $\mu(\cdot, A) = \infty \iff \nu(A) = \infty$.

From now on, **we suppress ω, that is, the first argument of μ**. Informally, one may think of the above formal definition as

$$\omega \mapsto \mu_\omega = \sum_i \delta_{u_i(\omega)}.$$

The deterministic measure ν is called the intensity measure of μ; that is, it gives the expected number of points falling in A.

Random measures provide enormous flexibility in modelling. A small sample of possible applications is as follows (we shall use the notation $\mathbb{N} = \{0, 1, 2, \cdots\}$):

(1) failure of electronic devices over time, $\mathbb{U} = \mathbb{R}^+$; one may also add the incurred damage, $\mathbb{U} = \mathbb{R}^+ \times \mathbb{R}^+$;

(2) the number of insects, spiders, or animals in a given field or region, $\mathbb{U} \subseteq \mathbb{R}^2(\mathbb{S}^3)$; one may also distinguish by species, for example by letting $\mathbb{X}$ be the set of species names, so that $\mathbb{U} = \mathbb{R}^2 \times \mathbb{X}$;

(3) the number of goals in a footballer's career, $\mathbb{U} = \mathbb{R}^+ \times \{1\}$; the number of goals in a given football match, $\mathbb{U} = [0, 90] \times \mathbb{N}$; the number of goals with the team also recorded, $\mathbb{U} = [0, 90] \times \mathbb{N} \times \mathbb{N}$;

(4) road accidents, $\mathbb{U} = \mathbb{R}^+ \times \mathbb{N}$; road accidents with the associated current damage, $\mathbb{U} = \mathbb{R}^+ \times \mathbb{N} \times \mathbb{R}^+$;

(5) stars in the Universe, $\mathbb{U} = \mathbb{R}^3$;

(6) customer arrivals at a shopping centre with accompanying profit or loss, $\mathbb{U} = \mathbb{R}^+ \times \mathbb{R}$; with additional differentiation by sex, $\mathbb{U} = \mathbb{R}^+ \times \mathbb{R} \times \{0, 1\}$.

Here, we collect some standard formulae for Poisson random measures, following Kyprianou [9, Ch. 2].

Theorem 2.1. *Let μ be a Poisson random measure on $(\mathbb{U}, \mathcal{U}, \nu)$, and let $f : \mathbb{U} \to \mathbb{R}$ be measurable. Define*

$$X = \int_{\mathbb{U}} f(x)\mu(dx). \tag{2.1}$$

Then the following hold:

(1) the integral X is almost surely absolutely convergent if and only if

$$\int_{\mathbb{U}} (1 \wedge |f(x)|)\, \nu(dx) < \infty; \tag{2.2}$$

(2) whenever X is almost surely finite,

$$\mathbb{E}\left[e^{zX}\right] = e^{\int_{\mathbb{U}} \left(e^{zf(x)} - 1\right)\nu(dx)}; \tag{2.3}$$

at least for $z \in i\mathbb{R}$ and extends to the domain of analyticity of $\int_{\mathbb{U}} \left(e^{zf(x)} - 1\right) \nu(dx)$.

(3) if, in addition,

$$\int_{\mathbb{U}} |f(x)|\nu(dx) < \infty,$$

then

$$\mathbb{E}[X] = \int_{\mathbb{U}} f(x)\nu(dx); \tag{2.4}$$

(4) if, moreover,

$$\int_{\mathbb{U}} f(x)^2\nu(dx) < \infty \qquad \text{and} \qquad \int_{\mathbb{U}} |f(x)|\nu(dx) < \infty,$$

then

$$\mathbb{E}\left[X^2\right] = \int_{\mathbb{U}} f(x)^2\nu(dx) + \left(\int_{\mathbb{U}} f(x)\nu(dx)\right)^2. \tag{2.5}$$

If $\mathbb{U} = \mathbb{R}^+ \times \{1\} \cong \mathbb{R}^+$, then the Poisson random measure is called a Poisson counting measure, and its points may be interpreted as event times or arrivals on the time axis. For the moment, assume that

$$\nu(B) = \mathbb{E}\left[\mu(B)\right] < \infty$$

for every bounded Borel set $B \subseteq \mathbb{R}^+$. For $t \geq 0$, set

$$N_t := \mu\left([0,t]\right) \sim Poi\left(\nu\left([0,t]\right)\right).$$

Thus N_t counts the number of points in $[0,t]$. Since

$$N_{t+s} - N_t = \mu\left((t, t+s]\right),$$

and $[0,t] \cap (t, t+s] = \emptyset$, the defining independence property of the Poisson random measure gives independence of increments, i.e. for $t, s > 0$,

$$N_t \perp\!\!\!\perp N_{t+s} - N_t.$$

Definition 2.1. *A Poisson counting measure defines a Poisson counting process with rate $\lambda > 0$ if, for every interval $(a,b) \subset \mathbb{R}^+$,*

$$\nu((a,b)) = \lambda(b-a).$$

Then

$$N = \left(N_t\right)_{t\geq 0}, \qquad N_t = \mu\left([0,t]\right), \qquad t \geq 0,$$

is called a Poisson counting process with rate λ.

Remark 2.2. *Since single points are not charged by Lebesgue measure, open, closed, or half-open intervals may be used interchangeably in the definition above. Moreover, the expected number of points in (a,b) is proportional to $b-a$.*

From Definition 2.1, it follows that $N = \left(N_t\right)_{t\geq 0}$ has independent and stationary increments. Indeed, note that, for $t, s > 0$,

$$N_{t+s} - N_t \sim Poi(\lambda s) \sim N_s.$$

For such N, we define a compound Poisson process by

$$X = (X_t)_{t\geq 0} = \left(\sum_{j=1}^{N_t} \xi_j\right)_{t\geq 0},$$

where $\left(\xi_j\right)_{j\geq 1}$ are i.i.d. and independent of N. We adopt the convention $\sum_{j=1}^{0} = 0$.

Let us recall the characteristic function formula, which captures much of the information about the process and corresponds to (2.3) in Theorem 2.1:

$$\mathbb{E}\left[e^{i\zeta X_t}\right] = e^{\lambda t \int \left(e^{i\zeta x} - 1\right) F(dx)} = e^{t \int \left(e^{i\zeta x} - 1\right) \Pi(dx)} = e^{t\Psi(i\zeta)}, \qquad \zeta \in \mathbb{R}, \tag{2.6}$$

where $F(dx) = \mathbb{P}(\xi_1 \in dx)$. The quantities Π and Ψ are the input data; in principle, any property of the process that can be read from them is regarded as theoretically understood.

In the language of random measures, this means that

$$\mathbb{U} = \mathbb{R}^+ \times \mathbb{R}^d,$$

usually with $d = 1$, and

$$\nu(A) = \lambda \int_A dt\, F(dx) = \lambda \int_A dt\, \mathbb{P}(\xi_1 \in dx), \qquad A \in \mathcal{B}(\mathbb{U}).$$

Equivalently, we write

$$\nu(dt, dx) = \underbrace{\lambda\, dt}_{N}\, \underbrace{F(dx)}_{\xi_1} = dt\, \underbrace{\Pi(dx)}_{\lambda F(dx)}. \tag{2.7}$$

This notation separates the arrival rate from the jump-size distribution.

Finally, a compound Poisson process is a continuous-time random walk: its jumps form a random walk, while the waiting times are i.i.d. exponential random variables with parameter λ. Hence standard asymptotic results such as the Law of Large Numbers (LLN), Central Limit Theorem (CLT), and Law of the Iterated Logarithm (LIL) also apply.

3. Lévy processes : definition and basic properties

The probability space $\{\Omega, \mathcal{F}, \mathbb{P}\}$ is chosen so that $\Omega = \mathcal{D}(\mathbb{R}^+)$, $\mathcal{F}$ is the completion of the sigma-algebra[1] generated by the Skorokhod topology on $\mathcal{D}(\mathbb{R}^+)$, and $\mathbb{P}$ is a probability measure on $\mathcal{F}$; see [4] for the definition of the Skorokhod topology.[2] For a stochastic process $X = (X_t)_{t\geq 0}$, we consider the natural filtration given by the completed sigma-algebras

$$\mathcal{F}_t = \sigma(X_s : s \leq t), \qquad t \geq 0.$$

On this probability space, $X(\omega) = \omega$, $\omega \in \Omega$, with evaluations $X_t(\omega) = \omega(t)$, $t \geq 0$.

Definition 3.1. *A stochastic process $X = (X_t)_{t\geq 0}$ with $X_0 = 0$ is a Lévy process on $\{\Omega, \mathcal{F}, \mathbb{P}\}$ if, with respect to the filtration $(\mathcal{F}_t)_{t\geq 0}$, it has the following properties:*

(1) for any $t, s > 0$, $X_{t+s} - X_t \overset{d}{=} X_s$ — stationary increments;
(2) for any $t, s > 0$, $X_{t+s} - X_t \perp\!\!\!\perp \mathcal{F}_t$ — independent increments;

[1]Let $\mathcal{A}$ be a sigma-algebra and $\mathcal{N} = \{B : B \subseteq A, A \in \mathcal{A}, \mathbb{P}(A) = 0\}$; $\mathcal{A}_c = \{A \cup B : A \in \mathcal{A}, B \in \mathcal{N}\}$ is the completion.

[2]On $\mathcal{D}[0,1]$, the Skorokhod J_1-topology is defined by the metric

$$d(x,y) = \inf_{\lambda\in\Lambda} \max\{\sup_{0\leq t\leq 1} |\lambda(t) - t|,\ \sup_{0\leq t\leq 1} |x(t) - y(\lambda(t))|\},$$

where Λ is the set of all increasing homeomorphisms of $[0,1]$. This makes $\mathcal{D}[0,1]$ a separable metric space. The corresponding topology on $\mathcal{D}(\mathbb{R}^+)$ is obtained by a standard compactification of time.

(3) it has almost surely right-continuous paths with left limits (rcll).

Remark 3.1. *The usual requirement that X has rcll paths is already built into the choice of Ω. More generally, if a process has stationary and independent increments and is continuous in probability at zero, that is,*

$$\lim_{\epsilon\to 0} \mathbb{P}\left(|X_\epsilon| > x\right) = 0, \qquad x > 0,$$

then it admits a modification with right-continuous paths with left limits, independently of the underlying probability space. One way to obtain this is via martingale regularisation; see, for example, Kallenberg, Theorem 7.27(ii) in the first edition, when the filtration $(\mathcal{F}_t)_{t\geq 0}$ is right-continuous. This can be verified using the monotone class theorem together with an approximation argument; see also blog post.

Remark 3.2. *For Lévy processes, the natural completed filtration $(\mathcal{F}_t)_{t\geq 0}$ is right-continuous, that is,*

$$\mathcal{F}_t = \bigcap_{s>t} \mathcal{F}_s, \qquad t \geq 0.$$

For $t = 0$, this follows from the observation that the path $(X_t)_{t\leq 1}$ can be reconstructed from the increments

$$\eta_n := \left(X_{2^{-n}+s} - X_{2^{-n}}\right)_{s\leq 2^{-n}}, \qquad n \geq 1.$$

These path segments are mutually independent, and

$$\sigma\left(\eta_k,\ k \geq n\right) = \mathcal{F}_{2^{-n}}.$$

Hence, by Kolmogorov's 0-1 law, the tail sigma-algebra $\bigcap_{n\geq 1} \sigma\left(\eta_k,\ k\geq n\right) = \bigcap_{n\geq 1} \mathcal{F}_{2^{-n}}$ is $\mathbb{P}$-trivial, and coincides with $\mathcal{F}_0$. For general $t \geq 0$, right-continuity follows from the independence of increments and the decomposition

$$\mathcal{F}_{t+h} = \sigma\left(\mathcal{F}_t, \mathcal{F}'_h\right), \qquad \mathcal{F}'_h = \sigma\left(X_{t+s} - X_t,\ s \leq h\right),$$

together with the triviality of $\bigcap_{h>0} \mathcal{F}'_h$.

The random variable X_1 is ***infinitely divisible***. Indeed, for every $n \geq 1$,

$$X_1 = \sum_{j=1}^{n} \left(X_{\frac{j}{n}} - X_{\frac{j-1}{n}}\right), \tag{3.1}$$

and the increments $\left(X_{\frac{j}{n}} - X_{\frac{j-1}{n}}\right)_{j=1}^{n}$ are i.i.d.

For random variables ξ and η, independence is equivalent to the factorisation

$$\mathbb{E}\left[f(\xi)g(\eta)\right] = \mathbb{E}\left[f(\xi)\right]\mathbb{E}\left[g(\eta)\right]$$

for all bounded measurable functions f, g. Equivalently, in terms of characteristic functions,

$$\mathbb{E}\left[e^{i\theta_1\xi+i\theta_2\eta}\right] = \mathbb{E}\left[e^{i\theta_1\xi}\right]\mathbb{E}\left[e^{i\theta_2\eta}\right], \qquad \theta_1, \theta_2 \in \mathbb{R}.$$

In many arguments below, it will be convenient to verify independence on a generating class. For example, it is enough to check

$$\mathbb{P}\left(\xi \in A, \eta \in B\right) = \mathbb{P}\left(\xi \in A\right)\mathbb{P}\left(\eta \in B\right)$$

for all A, B belonging to π-systems that generate $\mathcal{B}(\mathbb{R})$. One may take, for instance, intervals of the form $(-\infty, t]$, half-open intervals, or open intervals; the choice is immaterial by the π-λ theorem.

3.1. **The jump process and the process of left limits.** For a Lévy process $(X_t)_{t\geq 0}$, the left limits

$$X_{t-} := \lim_{\epsilon\to 0+} X_{t-\epsilon}$$

exist simultaneously almost surely for all $t > 0$. We write

$$X_- = (X_{t-})_{t\geq 0}, \qquad X_{0-} = X_0,$$

for the process of left limits. Defining

$$\Delta_t = X_t - X_{t-}, \qquad t \geq 0,$$

we obtain the jump process

$$\Delta = (\Delta_t)_{t\geq 0}\,. \tag{3.2}$$

Thus

$$X = X_- + \Delta. \tag{3.3}$$

Let

$$\mathcal{F}_{t-} = \sigma\left(\bigcup_{s<t}\mathcal{F}_s\right), \qquad t > 0, \qquad \mathcal{F}_{0-} = \mathcal{F}_0. \tag{3.4}$$

The processes X_- and Δ are adapted to $(\mathcal{F}_{t-})_{t\geq 0}$, and to $(\mathcal{F}_t)_{t\geq 0}$, respectively. Equivalently, for every $t \geq 0$, $(X_{s-})_{s\leq t}$ and $(\Delta_s)_{s\leq t}$ are $\mathcal{F}_{t-}$- and $\mathcal{F}_t$-measurable, respectively. The jumps of each path are at most countable; hence $\Delta_t = 0$ for all t, except at countably many jump times.

3.2. **Strong Markov property and martingale property.** We discuss a basic property that will be used repeatedly below. We assume the right-continuity of the underlying filtration.

A mapping $\tau : \Omega \to \mathbb{R}^+$ is a stopping time if

$$\{\tau \leq t\} \in \mathcal{F}_t, \qquad t \geq 0.$$

Associated with such a τ, the stopped sigma-algebra is defined by

$$\mathcal{F}_\tau = \{A \in \mathcal{F} : \forall t \geq 0,\ A \cap \{\tau \leq t\} \in \mathcal{F}_t\}. \tag{3.5}$$

Thus a stopping time is a random time whose occurrence up to time t can be decided from the information available at time t. We shall use the following standard result.

Lemma 3.3. *Let τ, σ be stopping times. Then:*

(1) $\mathcal{F}_\tau$ is a sigma-algebra;
(2) if $\tau \leq \sigma$, then $\mathcal{F}_\tau \subseteq \mathcal{F}_\sigma$;
(3) τ is $\mathcal{F}_\tau$-measurable;
(4) if $X = (X_t)_{t \geq 0}$ is adapted and right-continuous, then $X_\tau 1_{\tau < \infty}$ is $\mathcal{F}_\tau$-measurable.

Proof. The first claim follows directly from the definition. If $A \in \mathcal{F}_\tau$, then $A^c \in \mathcal{F}_\tau$ since

$$A^c \cap \{\tau \leq t\} = \{\tau \leq t\} \setminus (A \cap \{\tau \leq t\}) \in \mathcal{F}_t.$$

Also, if $A_n \in \mathcal{F}_\tau, n \geq 1$, then

$$\left(\bigcup_{n \geq 1} A_n\right) \cap \{\tau \leq t\} = \bigcup_{n \geq 1} (A_n \cap \{\tau \leq t\}) \in \mathcal{F}_t.$$

Hence $\mathcal{F}_\tau$ is a sigma-algebra. For the second claim, let $A \in \mathcal{F}_\tau$. If $\tau \leq \sigma$, then

$$A \cap \{\sigma \leq t\} = A \cap \{\tau \leq t\} \cap \{\sigma \leq t\} \in \mathcal{F}_t.$$

Therefore $A \in \mathcal{F}_\sigma$. The third claim is clear. For the final claim, fix $t \geq 0$, and set

$$\tau_n = 2^{-n} \lceil 2^n \tau \rceil.$$

Then $\tau_n \downarrow \tau$ and, on $\{\tau \leq t\}$, we have $\tau_n \leq t + 2^{-n}$. Hence

$$X_{\tau_n} 1_{\tau \leq t} \text{ is } \mathcal{F}_{t+2^{-n}} \text{ measurable.}$$

By right-continuity of X,

$$X_{\tau_n} 1_{\tau \leq t} \to X_\tau 1_{\tau \leq t}.$$

Thus $X_\tau 1_{\tau \leq t}$ is $\bigcap_{m \geq 1} \mathcal{F}_{t+1/m} = \mathcal{F}_t$ measurable. If B is a Borel set then

$$1_B(X_\tau 1_{\tau < \infty}) 1_{\tau \leq t} = 1_{B \setminus \{0\}}(X_\tau 1_{\tau \leq t}) + 1_{\tau \leq t} - 1_{\mathbb{R} \setminus \{0\}}(X_\tau 1_{\tau \leq t}) \in \mathcal{F}_t.$$

Therefore, for every Borel set B, $\{X_\tau \in B\} \cap \{\tau \le t\} \in \mathcal{F}_t$, which means that $X_\tau 1_{\tau<\infty}$ is $\mathcal{F}_\tau$-measurable. $\square$

For Lévy processes, the strong Markov property holds, extending stationary and independent increments to stopping times. To state it, for any stopping time τ, on $\{\tau < \infty\}$, we introduce the shifted processes

$$X^{(\tau)} = \left(X_t^{(\tau)}\right)_{t\ge 0} = (X_{\tau+t} - X_\tau)_{t\ge 0}\,, \qquad \Delta^{(\tau)} = \left(\Delta_t^{(\tau)}\right)_{t\ge 0} = (\Delta_{\tau+t})_{t\ge 0}\,. \tag{3.6}$$

We then have the following result.

Theorem 3.4. *Let X be a Lévy process and let τ be a stopping time. Then, conditionally on $\{\tau < \infty\}$, we have*

$$X^{(\tau)} \perp\!\!\!\perp (X_s)_{s\le\tau} \quad \text{and} \quad X^{(\tau)} \overset{w}{=} X. \tag{3.7}$$

If $\mathbb{P}(\tau < \infty) = 1$, then (3.7) *holds unconditionally.*

Proof. For $n \ge 1$, on $\{\tau < \infty\}$, set

$$\tau_n := \sum_{k\ge 0} (k+1)2^{-n} 1_{\tau\in[k2^{-n},(k+1)2^{-n})}.$$

Then $\tau_n \ge \tau$, $\tau_n \downarrow \tau$, and τ_n is a stopping time, since for $t \in [k2^{-n}, (k+1)2^{-n})$,

$$\{\tau_n \le t\} = \{\tau \le k2^{-n}\} \in \mathcal{F}_{k2^{-n}} \subseteq \mathcal{F}_t.$$

Fix $s > 0$, let f be bounded and continuous, and let $A \in \mathcal{F}_\tau$. Since

$$A \cap \{\tau \in \left[k2^{-n}, (k+1)2^{-n}\right)\} \in \mathcal{F}_{(k+1)2^{-n}},$$

stationarity and independence of increments give

$$\begin{aligned}
&\mathbb{E}\left[1_A 1_{\tau<\infty} f\left(X_{\tau_n+s} - X_{\tau_n}\right)\right] \\
&= \sum_{k\ge 0} \mathbb{E}\left[1_A 1_{\tau\in[k2^{-n},(k+1)2^{-n})} f\left(X_{(k+1)2^{-n}+s} - X_{(k+1)2^{-n}}\right)\right] \\
&= \mathbb{E}\left[f(X_s)\right] \sum_{k\ge 0} \mathbb{P}\left(A \cap \{\tau \in \left[k2^{-n}, (k+1)2^{-n}\right)\}\right) = \mathbb{E}\left[f(X_s)\right] \mathbb{P}\left(A \cap \{\tau < \infty\}\right).
\end{aligned}$$

Letting $n \to \infty$, right-continuity and dominated convergence yield

$$\mathbb{E}\left[1_A 1_{\tau<\infty} f\left(X_{\tau+s} - X_\tau\right)\right] = \mathbb{E}\left[f(X_s)\right] \mathbb{P}\left(A \cap \{\tau < \infty\}\right).$$

Thus, on $\{\tau < \infty\}$, $X_{\tau+s} - X_\tau$ is independent of $\mathcal{F}_\tau$ and has the same law as X_s. The same argument applies to

$$\left(X_{\tau+t_1} - X_\tau, \ldots, X_{\tau+t_m} - X_\tau\right).$$

Hence

$$X^{(\tau)} = (X_{\tau+t} - X_\tau)_{t\geq 0} \overset{w}{=} (X_t)_{t\geq 0} \quad \text{and} \quad X^{(\tau)} = (X_{\tau+t} - X_\tau)_{t\geq 0} \perp\!\!\!\perp \mathcal{F}_\tau$$

on $\{\tau < \infty\}$. Since $(X_s)_{s\leq\tau}$ is $\mathcal{F}_\tau$-measurable, this implies (3.7). If $\mathbb{P}(\tau < \infty) = 1$, the indicators $1_{\tau<\infty}$ may be omitted. □

The last result shows that a Lévy process possesses the strong Markov property. In particular, for every stopping time τ and every bounded measurable function f,

$$\mathbb{E}\left[f\left(X_{\tau+t}\right)1_{\tau<\infty}\right] = \int_{\mathbb{R}^d} \mathbb{E}\left[f\left(X_t + y\right)\right]\mathbb{P}\left(X_\tau \in dy, \tau < \infty\right).$$

Equivalently, for bounded measurable functions f, g, Theorem 3.4 gives

$$\mathbb{E}\left[f\left(X_{\tau+t} - X_\tau\right) g\left(X_\tau\right) \mid \tau < \infty\right] = \mathbb{E}\left[f\left(X_t\right)\right]\mathbb{E}\left[g\left(X_\tau\right) \mid \tau < \infty\right]. \tag{3.8}$$

By the strong Markov property, on $\{\tau < \infty\}$ we also have

$$\left(X^{(\tau)}, \Delta^{(\tau)}\right) \overset{w}{=} (X, \Delta), \quad \left(X^{(\tau)}, \Delta^{(\tau)}\right) \perp\!\!\!\perp_{\{\tau<\infty\}} \mathcal{F}_\tau, \tag{3.9}$$

which is simply a consequence of the identity $\Delta = X - X_-$, since X_- is determined pathwise by X.

4. Characterisation of Lévy processes

In this section, we characterise the form of a general Lévy process and derive the main formula for the characteristic function of X_t, namely the Lévy-Khintchine exponent. We begin with two classical examples.

Example: The characteristic function of Brownian motion with linear drift is

$$\begin{aligned} \mathbb{E}\left[e^{i\zeta(\sigma B_t + \mu t)}\right] &= e^{i\zeta\mu t}e^{-t\frac{\sigma^2\zeta^2}{2}} = e^{t\Psi(i\zeta)}, \\ \Psi(i\zeta) &= \mu i\zeta + \frac{\sigma^2}{2}(i\zeta)^2, \qquad \zeta \in \mathbb{R}. \end{aligned} \tag{4.1}$$

Note that Ψ extends to an entire function in the complex plane.

Example: Let $X_t = \sum_{j=1}^{N_t} \xi_j$ be a compound Poisson process with N a Poisson process of rate λ, and with $(\xi_j)_{j\geq 1}$ i.i.d. with law F, independent of N. Then

$$\begin{aligned} \mathbb{E}\left[e^{i\zeta X_t}\right] &= \mathbb{E}\left[\left(\mathbb{E}\left[e^{i\zeta\xi_1}\right]\right)^{N_t}\right] = \exp\left(\lambda t\left(\mathbb{E}\left[e^{i\zeta\xi_1}\right] - 1\right)\right) \\ &= \exp\left(t\int\left(e^{i\zeta x} - 1\right)\Pi(dx)\right) = e^{t\Psi(i\zeta)}, \qquad \zeta \in \mathbb{R}, \end{aligned} \tag{4.2}$$

where $\Pi(dx) = \lambda F(dx)$. This coincides with (2.6).

Moreover, X has stationary and independent increments. Indeed, for $0 \le s < t$,

$$X_t - X_s = \sum_{j=N_s+1}^{N_t} \xi_j = \sum_{j=1}^{N_t - N_s} \xi_{j+N_s}.$$

Since $N_t - N_s \perp\!\!\!\perp N_s$, $N_t - N_s \sim Poi(\lambda(t-s))$, and the unused jump sizes are independent of $\mathcal{F}_s$, we have

$$X_t - X_s \overset{d}{=} \sum_{j=1}^{N_{t-s}} \xi_j = X_{t-s}$$

and $X_t - X_s \perp\!\!\!\perp \mathcal{F}_s$. Since its paths are right-continuous with left limits, X is a Lévy process.

We begin with Lévy processes having finitely many jumps on every finite time interval.

4.1. **Finite number of jumps.** We start with an auxiliary lemma. Throughout the lemma, we assume that jumps are necessarily present.

Lemma 4.1. *Let $X = (X_t)_{t\ge 0}$ be a Lévy process with respect to the filtration $(\mathcal{F}_t)_{t\ge 0}$, with almost surely finitely many jumps on every finite interval $[a,b]$, $0 \le a < b < \infty$. Let $T_0 = 0$,*

$$T_1 = \inf\{t > 0 : \Delta_t \neq 0\}, \qquad T_{j+1} = \inf\{t > T_j : \Delta_t \neq 0\}, \qquad j \ge 0,$$

and $\mathbb{P}(T_1 < 1) > 0$. Then $\mathbb{P}(T_j < \infty) = 1$ for all $j \ge 1$. With

$$\xi_j = X_{T_j} - X_{T_j-}, \qquad j \ge 1,$$

the random vectors $((T_j - T_{j-1}, \xi_j))_{j\ge 1}$ are independent and identically distributed, with $T_1 \sim Exp(\lambda)$ for some $\lambda > 0$. Furthermore,

$$\xi_j \perp\!\!\!\perp T_j - T_{j-1}, \qquad j \ge 1.$$

As a result,

$$Z_t = \sum_{j=1}^{N_t} \xi_j = \sum_{s\le t} \Delta_s, \qquad t \ge 0,$$

with

$$N_t = \sup\{j \ge 0 : T_j \le t\}, \qquad t \ge 0,$$

is a compound Poisson process.

Proof. T_1 is a stopping time, since for any $t > 0$,

$$\{T_1 \le t\} = \{\exists\, 0 < s \le t : \Delta_s \neq 0\} \in \mathcal{F}_t.$$

We next show that $T_1 \sim Exp(\lambda)$ for some $\lambda > 0$. Let $t > s > 0$, and set

$$A = \{T_1 > s\}, \qquad B = \{\Delta_v^{(s)} = 0,\ 0 < v \le t-s\} = \{T_1^{(s)} > t-s\}.$$

Then $\{T_1 > t\} = A \cap B$. The event B depends only on $X^{(s)}$, which, by (3.7) with $\tau = s$, is independent of $(X_v)_{v \le s}$, and by (3.9) has the same law as X. Hence A and B are independent, and

$$\mathbb{P}(T_1 > t \mid T_1 > s) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \mathbb{P}(B) = \mathbb{P}(T_1 > t - s).$$

Therefore $g(t) := \mathbb{P}(T_1 > t)$ satisfies

$$g(t) = g(t-s)g(s), \qquad g(0) = \mathbb{P}(T_1 > 0) = 1.$$

Since g is non-increasing, $g(t) = e^{-\lambda t}$ for some $\lambda \ge 0$. The assumption $\mathbb{P}(T_1 < \infty) = 1$ excludes $\lambda = 0$, and therefore $\lambda > 0$. Thus

$$T_1 \sim Exp(\lambda).$$

Next, note that

$$T_2 = \inf\{t > T_1 : \Delta X_t \neq 0\} = T_1 + \inf\{t > 0 : \Delta_t^{(T_1)} \neq 0\}.$$

By the strong Markov property, see (3.9), and since $\mathbb{P}(T_1 < \infty) = 1$,

$$T_2 - T_1 = \inf\{t > 0 : \Delta_t^{(T_1)} \neq 0\} \perp\!\!\!\perp \mathcal{F}_{T_1}, \qquad T_2 - T_1 \stackrel{d}{=} T_1 \sim Exp(\lambda).$$

Moreover, with

$$\xi_1 = X_{T_1} - X_{T_1-}, \qquad \xi_2 = X_{T_2} - X_{T_2-} = \xi_1^{(T_1)}.$$

ξ_1 is $\mathcal{F}_{T_1}$-measurable. Hence, by the strong Markov property,

$$(T_1, \xi_1) \stackrel{d}{=} (T_2 - T_1, \xi_2) \quad \text{and} \quad (T_1, \xi_1) \perp\!\!\!\perp (T_2 - T_1, \xi_2).$$

Repeating the argument shows that, if T_n is the nth jump time and

$$\xi_n = X_{T_n} - X_{T_n-}, \qquad n \ge 2,$$

then

$$(T_1, \xi_1),\ (T_2 - T_1, \xi_2),\ (T_3 - T_2, \xi_3), \ldots$$

are independent and identically distributed random vectors.

It remains to show that ξ_j is independent of $T_j - T_{j-1}$. We prove this only for $j = 1$, since the general case is identical. Let $a < b$ and $t > 0$. Denote by $\xi_1^{(t)}$ the first jump of $X^{(t)}$. On $\{T_1 > t\}$, we have $\xi_1 = \xi_1^{(t)}$, and therefore

$$\{\xi_1 \in (a,b)\} \cap \{T_1 > t\} = \{\xi_1^{(t)} \in (a,b)\} \cap \{T_1 > t\}.$$

Again by (3.9), $X^{(t)} \perp\!\!\!\perp (X_s)_{s\le t}$ and $X^{(t)} \overset{w}{=} X$. Hence

$$\begin{aligned}\mathbb{P}\left(\xi_1 \in (a,b)\,; T_1 > t\right) &= \mathbb{P}\left(T_1 > t\right)\mathbb{P}\left(\xi_1^{(t)} \in (a,b) \mid T_1 > t\right)\\ &= \mathbb{P}\left(T_1 > t\right)\mathbb{P}\left(\xi_1^{(t)} \in (a,b)\right)\\ &= \mathbb{P}\left(T_1 > t\right)\mathbb{P}\left(\xi_1 \in (a,b)\right).\end{aligned}$$

Since $a < b$ and $t > 0$ are arbitrary,

$$\xi_1 \perp\!\!\!\perp T_1.$$

Thus the jump time is independent of the jump size. Now define

$$N_t = \sup\{j \ge 0 : T_j \le t\}.$$

Since the increments $T_j - T_{j-1}$ are independent and exponentially distributed with parameter λ, $N = (N_t)_{t\ge 0}$ is a Poisson counting process. Alternatively, this could be seen from the identity

$$\{N_t = n\} = \{\{\# s \le t :\, \Delta_s \ne 0\} = n\}, \qquad n \ge 0,\, t \ge 0,$$

and the strong Markov property.

Since the i.i.d. pairs $((T_j - T_{j-1}, \xi_j))_{j\ge 1}$ have independent coordinates, the waiting times and jump sizes are independent. Hence, with

$$Z_t = \sum_{j=1}^{N_t} \xi_j, \qquad t \ge 0,$$

the process Z is a compound Poisson process. □

Proposition 4.2. *Let $X = (X_t)_{t\ge 0}$ be a Lévy process with respect to $(\mathcal{F}_t)_{t\ge 0}$, with finitely many jumps on every finite interval $[a,b]$, $0 \le a < b < \infty$, and constant between any two jumps. Then X is a compound Poisson process. In particular, for some $\lambda > 0$ and some distribution F,*

$$\begin{aligned}&\mathbb{E}\left[e^{i\theta X_t}\right] = e^{t\Psi(i\theta)}, \qquad \theta \in \mathbb{R},\\ &\Psi(i\theta) = \lambda \int_{-\infty}^{\infty}\left(e^{i\theta x} - 1\right) F(dx) = \int_{-\infty}^{\infty}\left(e^{i\theta x} - 1\right)\Pi(dx),\end{aligned} \tag{4.3}$$

where $\Pi(dx) = \lambda F(dx)$.

Proof. This follows immediately from Lemma 4.1, since in the present case $X = Z$. □

The next proposition allows the Lévy process to evolve continuously between its jumps.

Proposition 4.3. *If a Lévy process has continuous paths, then it is a Brownian motion with linear drift. More generally, if a Lévy process has finitely many jumps on every finite time interval, then*

$$X_t = \sigma B_t + \mu t + Z_t, \qquad t \geq 0,$$

where B is a Brownian motion and Z is an independent compound Poisson process. Its characteristic function is

$$\tag{4.4} \begin{aligned} \mathbb{E}\left[e^{i\theta X_t}\right] &= e^{t\left(\int\left(e^{i\theta x}-1\right)\Pi(dx)-\frac{\sigma^2\theta^2}{2}+i\mu\theta\right)}, \\ \Psi(i\theta) &= \int\left(e^{i\theta x}-1\right)\Pi(dx) - \frac{\sigma^2\theta^2}{2} + i\mu\theta, \qquad \theta \in \mathbb{R}, \end{aligned}$$

where $\Pi(dx) = \lambda F(dx)$ and F is the law of the jumps.

Proof. We prove the second statement directly; the first is the continuous case and was established earlier. Write

$$X_t = Y_t + \sum_{j=1}^{N_t} \xi_j = Y_t + Z_t.$$

By Lemma 4.1, Z is a compound Poisson process. We prove two claims.

1. Y is a Brownian motion with drift.

Since $Z = (Z_t)_{t\geq 0}$ is determined by the jump process of X, it is adapted to the natural filtration, and hence so is $Y = X - Z$. Moreover,

$$Y^{(t)} = X^{(t)} - Z^{(t)}.$$

By (3.9) $(X, \Delta) \overset{w}{=} (X^{(t)}, \Delta^{(t)})$ with $(X^{(t)}, \Delta^{(t)}) \perp\!\!\!\perp \mathcal{F}_t$. Hence $Y^{(t)} \perp\!\!\!\perp \mathcal{F}_t$ and

$$Y_s^{(t)} = X_s^{(t)} - \sum_{v\leq s} \Delta_v^{(t)} \overset{d}{=} X_s - \sum_{v\leq s} \Delta_v = Y_s.$$

Thus Y has stationary and independent increments. Since all jumps of X have been removed by Z, the process Y has continuous paths. Therefore

$$Y_t = \sigma B_t + \mu t.$$

2. Y and Z are independent.

It is enough to prove the two-parameter factorisation of the joint characteristic function. We do this for $t = 1$; the general case is identical. Let $\alpha, \beta \in \mathbb{R}$, and set

$$h(\alpha, \beta, n) = 1 - \mathbb{E}\left[e^{i\alpha Y_{1/n} + i\beta Z_{1/n}}\right].$$

Note that $\lim_{n\to\infty} h(\alpha,\beta,n)=0$ since $X_{\frac{1}{n}}\to 0$ a.s. Since $(Y_{\frac{j}{n}}-Y_{\frac{j-1}{n}}, Z_{\frac{j}{n}}-Z_{\frac{j-1}{n}})_{j\le n}, n\geq 1$ are i.i.d., we have

$$\begin{aligned}\mathbb{E}\left[e^{i\alpha Y_1+i\beta Z_1}\right] &= \lim_{n\to\infty}\left(\mathbb{E}\left[e^{i\alpha Y_{1/n}+i\beta Z_{1/n}}\right]\right)^n \\ &= \lim_{n\to\infty} e^{(n\log(1-h(\alpha,\beta,n)))} = e^{-\lim_{n\to\infty} nh(\alpha,\beta,n)},\end{aligned}$$

where we use $h(\alpha,\beta,n)\to 0$ and $\log(1-z)\sim -z$ as $z\to 0$. We compute $\lim_{n\to\infty} nh(\alpha,\beta,n)$. Since

$$1-e^{i\alpha Y_{1/n}+i\beta Z_{1/n}} = \left(1-e^{i\alpha Y_{1/n}}\right)+\left(1-e^{i\beta Z_{1/n}}\right)-\left(1-e^{i\alpha Y_{1/n}}\right)\left(1-e^{i\beta Z_{1/n}}\right),$$

we check that the mixed term is negligible, namely

$$\lim_{n\to\infty} n\mathbb{E}\left[\left(1-e^{i\alpha Y_{1/n}}\right)\left(1-e^{i\beta Z_{1/n}}\right)\right]=0.$$

Indeed, since $Z_{1/n}=0$ on $\{N_{1/n}=0\}$, using $\left|1-e^{ix}\right|\le |x|$ and $\left|1-e^{ix}\right|\le 2$, we obtain

$$n\left|\mathbb{E}\left[\left(1-e^{i\alpha Y_{1/n}}\right)\left(1-e^{i\beta Z_{1/n}}\right)\right]\right|\le 2|\alpha| n\mathbb{E}\left[\left|Y_{1/n}\right| 1_{N_{1/n}\geq 1}\right].$$

For $\varepsilon>0$ we write

$$\mathbb{E}\left[\left|Y_{1/n}\right| 1_{N_{1/n}\ge 1}\right]=\mathbb{E}\left[\left|Y_{1/n}\right| 1_{\{|Y_{1/n}|\le\varepsilon\}}1_{N_{1/n}\ge 1}\right]+\mathbb{E}\left[\left|Y_{1/n}\right| 1_{\{|Y_{1/n}|>\varepsilon\}}1_{N_{1/n}\ge 1}\right].$$

The first term is bounded by

$$\mathbb{E}\left[\left|Y_{1/n}\right| 1_{\{|Y_{1/n}|\le\varepsilon\}}1_{N_{1/n}\ge 1}\right]\le\varepsilon\mathbb{P}\left(N_{1/n}\ge 1\right).$$

For the second term, we apply Hölder's inequality to get

$$\mathbb{E}\left[\left|Y_{1/n}\right| 1_{\{|Y_{1/n}|>\varepsilon\}}1_{N_{1/n}\ge 1}\right]\le\sqrt{\mathbb{E}\left[Y_{1/n}^2 1_{|Y_{1/n}|>\varepsilon}\right]}\sqrt{\mathbb{E}\left[1_{N_{1/n}\ge 1}\right]}.$$

Therefore

$$\mathbb{E}\left[\left|Y_{1/n}\right| 1_{N_{1/n}\ge 1}\right]\le\varepsilon\mathbb{P}\left(N_{1/n}\ge 1\right)+\sqrt{\mathbb{E}\left[Y_{1/n}^2 1_{|Y_{1/n}|>\varepsilon}\right]}\sqrt{\mathbb{P}\left(N_{1/n}\ge 1\right)}.$$

Since $\mathbb{P}\left(N_{1/n}\ge 1\right)=1-e^{-\frac{\lambda}{n}}\sim\lambda/n$, and since

$$n\mathbb{E}\left[Y_{1/n}^2 1_{|Y_{1/n}|>\varepsilon}\right]\to 0$$

by $Y_t=\mu t+\sigma B_t$ and the scaling of the normal law[3], it follows that

$$\limsup_{n\to\infty} n\left|\mathbb{E}\left[\left(1-e^{i\alpha Y_{1/n}}\right)\left(1-e^{i\beta Z_{1/n}}\right)\right]\right|\le 2|\alpha|\lambda\varepsilon.$$

Letting $\varepsilon\downarrow 0$ proves the claim.

[3]For large n, we have $1_{|Y_{1/n}|>\epsilon}\le 1_{\sigma|B_{1/n}|>\epsilon/2}\stackrel{d}{=}1_{\sigma|B_1|>\sqrt{n}\epsilon/2}$ and $Y_{1/n}^2\stackrel{d}{=}\mu^2 n^{-2}+2\sigma\mu n^{-3/2}B_1+\sigma^2 n^{-1}B_1^2$

Therefore

$$\lim_{n\to\infty} nh(\alpha,\beta,n) = \lim_{n\to\infty} n\left(1-\mathbb{E}\left[e^{i\alpha Y_{1/n}}\right]\right) + \lim_{n\to\infty} n\mathbb{E}\left[\left(1-e^{i\beta Z_{1/n}}\right)\right].$$

The first term is

$$\lim_{n\to\infty} n\left(1-e^{i\alpha\mu/n}e^{-\sigma^2\alpha^2/(2n)}\right) = \frac{\sigma^2\alpha^2}{2} - i\alpha\mu.$$

For the compound Poisson part,

$$\begin{aligned}
&\lim_{n\to\infty} n\mathbb{E}\left[\left(1-e^{i\beta Z_{1/n}}\right)\right] \\
&= \mathbb{E}\left[\left(1-e^{i\beta\xi_1}\right)\right]\lim_{n\to\infty} n\mathbb{P}\left(N_{1/n}=1\right) + \lim_{n\to\infty} n\mathbb{E}\left[\left(1-e^{i\beta Z_{1/n}}\right)1_{N_{1/n}\geq 2}\right] \\
&= \lambda\mathbb{E}\left[\left(1-e^{i\beta\xi_1}\right)\right],
\end{aligned}$$

where we have used $\mathbb{P}\left(N_{1/n}\geq 2\right) = 1-e^{-\frac{\lambda}{n}} - \frac{\lambda}{n}e^{-\frac{\lambda}{n}} = o(1/n)$. Hence

$$\lim_{n\to\infty} nh(\alpha,\beta,n) = \frac{\sigma^2\alpha^2}{2} - i\alpha\mu + \lambda\mathbb{E}\left[\left(1-e^{i\beta\xi_1}\right)\right],$$

and consequently

$$\begin{aligned}
\mathbb{E}\left[e^{i\alpha Y_1 + i\beta Z_1}\right] &= e^{-\frac{\sigma^2\alpha^2}{2}+i\alpha\mu}e^{-\lambda\mathbb{E}\left[\left(1-e^{i\beta\xi_1}\right)\right]} \\
&= \mathbb{E}\left[e^{i\alpha Y_1}\right]\mathbb{E}\left[e^{i\beta Z_1}\right].
\end{aligned}$$

Thus $Y_1 \perp\!\!\!\perp Z_1$. Applying the same argument to increments over disjoint intervals gives independence of the processes Y and Z.

Finally,

$$\mathbb{E}\left[e^{i\theta X_t}\right] = \mathbb{E}\left[e^{i\theta Y_t}\right]\mathbb{E}\left[e^{i\theta Z_t}\right] = \exp\left(t\left(-\frac{\sigma^2\theta^2}{2} + i\mu\theta + \lambda\int\left(e^{i\theta x}-1\right)F(dx)\right)\right),$$

which is (4.4).

□

If we think of a process as consisting of a continuous part plus finitely many jumps, there is no way to enlarge the class of Lévy processes.

4.2. **Lévy processes with bounded variation.** The results of Subsection 4.1, and in particular Proposition 4.3, show that any Lévy process with finitely many jumps on every finite time interval can be represented as

$$X_t = \mu t + \sigma B_t + \sum_{j=1}^{N_t}\xi_j,$$

where the components are mutually independent. In terms of the jump process of X,

$$\sum_{j=1}^{N_t} \xi_j = \sum_{s \leq t} \Delta_s.$$

This raises the question: can Lévy processes have infinitely many jumps, that is, *infinite activity*, on a finite time interval?

Lemma 4.4. *Assume that*

$$X_t = \sum_{s \leq t} \Delta_s, \qquad \sum_{s \leq t} |\Delta_s| < \infty$$

almost surely for all $t > 0$, *and fix* $\epsilon > 0$. *Define* $T_0^\epsilon = 0$ *and, recursively,*

$$T_{j+1}^\epsilon = \inf\{t > T_j^\epsilon : |\Delta_t| > \epsilon\},$$

with the convention $\inf \emptyset = \infty$. *Also set*

$$\xi_j^\epsilon = \Delta_{T_j^\epsilon}, \qquad j \geq 1.$$

If $\mathbb{P}(T_1^\epsilon < 1) > 0$, *then*

$$X_t^\epsilon = \sum_{s \leq t} \Delta_s 1_{|\Delta_s| > \epsilon} = \sum_{j=1}^{N_t^\epsilon} \xi_j^\epsilon$$

is a compound Poisson process. Equivalently, it is generated by a Poisson random measure on $\mathbb{R}^+ \times \mathbb{R}$ *with intensity*

$$dt\, \Pi^\epsilon(dx) = dt\, \left(\mathbb{E}\left[T_1^\epsilon\right]\right)^{-1} \mathbb{P}\left(\xi_1^\epsilon \in dx\right),$$

supported on $\mathbb{R}^+ \times \{x \in \mathbb{R} : |x| > \epsilon\}$. *In particular,*

$$\Pi^\epsilon(\mathbb{R}) = \left(\mathbb{E}\left[T_1^\epsilon\right]\right)^{-1}.$$

Finally, if $\mathbb{P}(T_1^\epsilon < 1) = 0$ *then* $\Pi^\epsilon(dx)$ *is the zero measure on* $\{|x| > \epsilon\}$.

Proof. The proof is the same as that of Lemma 4.1, applied only to jumps whose absolute size exceeds ϵ. First, T_1^ϵ is a stopping time, since for any $t > 0$,

$$\{T_1^\epsilon \leq t\} = \{\exists\, 0 < s \leq t : |\Delta_s| > \epsilon\} \in \mathcal{F}_t.$$

Note that absolute convergence of $\sum_{s \leq t} \Delta_s$ implies that for any $\epsilon > 0$ the number of jumps larger than ϵ in absolute value is finite. Therefore, $T_1^\epsilon > 0$ can be defined properly. Let $t > s > 0$, and set

$$A = \{T_1^\epsilon > s\}, \qquad B = \{\text{no jumps with absolute value greater than } \epsilon \text{ on } (s,t]\}.$$

Then $\{T_1^\epsilon > t\} = A \cap B$. The event B depends only on the shifted process $X^{(s)}$, and hence is independent of $(X_v)_{v \leq s}$. Moreover, by stationarity of increments it has the same probability as $\{T_1^\epsilon > t - s\}$. Therefore

$$\mathbb{P}\left(T_1^\epsilon > t \mid T_1^\epsilon > s\right) = \mathbb{P}\left(T_1^\epsilon > t - s\right).$$

Thus $T_1^\epsilon \sim Exp(\lambda^\epsilon)$ for some $\lambda^\epsilon \geq 0$. Since $\mathbb{P}\left(T_1^\epsilon < 1\right) > 0$, we have $\lambda^\epsilon > 0$.

By the strong Markov property at T_j^ϵ, the vectors

$$\left((T_j^\epsilon - T_{j-1}^\epsilon, \xi_j^\epsilon)\right)_{j \geq 1}$$

are independent and identically distributed, with

$$T_j^\epsilon - T_{j-1}^\epsilon \sim Exp(\lambda^\epsilon).$$

The same argument as in Lemma 4.1 also gives

$$\xi_j^\epsilon \perp\!\!\!\perp T_j^\epsilon - T_{j-1}^\epsilon, \qquad j \geq 1.$$

Hence $N^\epsilon = (N_t^\epsilon)_{t \geq 0}$, where

$$N_t^\epsilon = \sup\{j \geq 0 : T_j^\epsilon \leq t\},$$

is a Poisson counting process with rate λ^ϵ, and

$$X_t^\epsilon = \sum_{j=1}^{N_t^\epsilon} \xi_j^\epsilon$$

is a compound Poisson process.

The associated Poisson random measure has intensity

$$dt\, \lambda^\epsilon \mathbb{P}\left(\xi_1^\epsilon \in dx\right).$$

Since $\mathbb{E}\left[T_1^\epsilon\right] = 1/\lambda^\epsilon$, this is exactly

$$dt\, \Pi^\epsilon(dx) = dt\ \left(\mathbb{E}\left[T_1^\epsilon\right]\right)^{-1} \mathbb{P}\left(\xi_1^\epsilon \in dx\right).$$

Finally,

$$\Pi^\epsilon\left(\mathbb{R}\right) = \lambda^\epsilon = \left(\mathbb{E}\left[T_1^\epsilon\right]\right)^{-1}.$$

$\square$

Lemma 4.5. *Let the assumptions and notation of Lemma 4.4 be in force. Then, with*

$$\Pi^\epsilon(dx) = \left(\mathbb{E}\left[T_1^\epsilon\right]\right)^{-1} \mathbb{P}\left(\xi_1^\epsilon \in dx\right),$$

for any $a > \epsilon$, the restriction of Π^ϵ to $\{|x| > a\}$ coincides with Π^a. Consequently, there exists a σ-finite measure Π on $\mathbb{R} \setminus \{0\}$ such that, for every $\epsilon > 0$,

$$\Pi(A) = \Pi^\epsilon(A), \qquad A \subseteq \{|x| > \epsilon\}, \quad A \in \mathcal{B}(\mathbb{R}).$$

Proof. Fix $a > 0$ and choose $0 < \epsilon < a$. By Lemma 4.4, X^a is a compound Poisson process generated by a Poisson random measure with intensity $dt\,\Pi^a(dx)$. On the other hand, thinning the jumps of X^ϵ gives

$$X_t^{a,\epsilon} := \sum_{j=1}^{N_t^\epsilon} \xi_j^\epsilon 1_{|\xi_j^\epsilon|>a}, \qquad t \geq 0.$$

With $f(x) = x1_{|x|>a}$ the exponential formula (2.3) yields

$$\mathbb{E}\left[e^{i\theta X_t^{a,\epsilon}}\right] = e^{t\int_{-\infty}^{\infty}(e^{i\theta f(x)}-1)\Pi^\epsilon(dx)} = e^{t\int_{|x|>a}(e^{i\theta x}-1)\Pi^\epsilon(dx)}$$

we check that $(X_t^{a,\epsilon})_{t\geq 0}$ is a compound Poisson process with intensity equal to the restriction of $dt\,\Pi^\epsilon(dx)$ to $\mathbb{R}^+ \times \{|x| > a\}$. Since $X^{a,\epsilon} = X^a$, the corresponding intensity measures agree, and hence

$$\Pi^\epsilon(dx)\big|_{\{|x|>a\}} = \Pi^a(dx).$$

In more detail, the latter can be checked from the fact that the expected number of jumps per unit time coincides for $X^{a,\epsilon}$ and X^a, i.e.

$$\Pi^\epsilon(\mathbb{R})\big|_{\{|x|>a\}} = \Pi^a(\mathbb{R}),$$

which together with

$$\int_{-\infty}^{\infty}\left(e^{i\theta f(x)} - 1\right)\Pi^\epsilon(dx) = \int_{-\infty}^{\infty}\left(e^{i\theta x} - 1\right)\Pi^a(dx)$$

and $f(x) = x1_{|x|>a}$ yield that the Fourier transforms of $\Pi^\epsilon(dx)\big|_{\{|x|>a\}}$ and $\Pi^a(dx)$ coincide.

This consistency allows us to define a measure Π on $\mathbb{R}\setminus\{0\}$ by setting

$$\Pi(A) := \Pi^\epsilon(A)$$

whenever $A \in \mathcal{B}(\mathbb{R})$ and $A \subseteq \{|x| > \epsilon\}$. The definition is independent of the choice of ϵ. Moreover, Π is σ-finite, since $\mathbb{R}\setminus\{0\} = \bigcup_{n=1}^\infty\{|x| > 1/n\}$, and

$$\Pi\left(\{|x| > 1/n\}\right) = \Pi^{1/n}\left(\mathbb{R}\right) < \infty, \qquad n \geq 1.$$

□

For a Lévy process of the form

$$X_t = \sum_{s\leq t}\Delta_s,$$

Lemma 4.5 shows that there exists a σ-finite measure $\Pi(dx)$ describing the intensity and distribution of its jumps in terms of a Poisson random measure. The next theorem gives a necessary and sufficient condition, in terms of Π, for the absolute convergence of $\sum_{s\leq t}\Delta_s$.

Theorem 4.6. *Let $\mu(ds, dx)$ be a Poisson random measure on $\mathbb{R}^+ \times \mathbb{R}$ with intensity $ds\,\Pi(dx)$. Then, almost surely,*

$$\sum_{s\le t} |\Delta_s| < \infty \text{ for every } t > 0 \iff \int_{\mathbb{R}} \min\{|x|, 1\}\Pi(dx) < \infty, \tag{4.5}$$

where Δ_s denotes the spatial coordinate of the point at time s.

Proof. For $a > 0$, set

$$\Delta_s^a = \Delta_s 1_{|\Delta_s|>a}.$$

This amounts to restricting the Poisson random measure to $\mathbb{R}^+ \times \{|x| > a\}$, with intensity

$$ds\,\Pi^a(dx) = ds\,\Pi(dx)1_{|x|>a}.$$

Then

$$Y_t^a = -\sum_{s\le t} |\Delta_s^a|$$

is a compound Poisson process. As $a \downarrow 0$, we have $Y_t^a \downarrow Y_t$, where

$$Y_t = -\sum_{s\le t} |\Delta_s| \in [-\infty, 0].$$

Consequently,

$$Y_t = -\infty \text{ a.s.} \iff \lim_{a\downarrow 0} \mathbb{E}\left[e^{Y_t^a}\right] = 0.$$

Since $Y_t^a \le 0, a > 0, t > 0$, then the exponential formula for Poisson random measures, see (2.3), extends for any z such that $\Re(z) \ge 0$ and thus for $z = 1$

$$\begin{aligned}\lim_{a\downarrow 0} \mathbb{E}\left[e^{Y_t^a}\right] &= \lim_{a\downarrow 0} \exp\left(t\int_{\mathbb{R}} \left(e^{-|x|} - 1\right) 1_{|x|>a}\Pi(dx)\right)\\ &= \exp\left(-t\int_{\mathbb{R}} \left(1 - e^{-|x|}\right) \Pi(dx)\right),\end{aligned}$$

where the second identity follows by monotone convergence. Hence

$$Y_t = -\infty \text{ a.s.} \iff \int_{\mathbb{R}} \left(1 - e^{-|x|}\right) \Pi(dx) = \infty.$$

It remains to compare this integral with

$$\int_{\mathbb{R}} \min\{|x|, 1\}\Pi(dx).$$

For $|x| \le 1$, there exists a constant $c > 0$ such that

$$c\,|x| \le 1 - e^{-|x|} \le |x|,$$

while for $|x| > 1$,

$$1 - e^{-1} \le 1 - e^{-|x|} \le 1.$$

Therefore

$$\int_{\mathbb{R}} \left(1 - e^{-|x|}\right) \Pi(dx) < \infty \iff \int_{\mathbb{R}} \min\{|x|, 1\}\Pi(dx) < \infty.$$

This proves the implication

$$\int_{\mathbb{R}} \min\{|x|, 1\}\Pi(dx) = \infty \implies \sum_{s \le t} |\Delta_s| = \infty \quad \text{a.s.}$$

Conversely, assume that

$$\int_{\mathbb{R}} \min\{|x|, 1\}\Pi(dx) < \infty.$$

Then

$$\mathbb{E}\left[e^{Y_t}\right] = \exp\left(-t \int_{\mathbb{R}} \left(1 - e^{-|x|}\right) \Pi(dx)\right) > 0,$$

and therefore

$$\mathbb{P}\left(Y_t > -\infty\right) > 0.$$

We now show that this probability is in fact equal to one. Put

$$q = \mathbb{P}\left(Y_t > -\infty\right).$$

For $\lambda > 0$, the exponential formula gives

$$\mathbb{E}\left[e^{\lambda Y_t}\right] = \exp\left(-t \int_{\mathbb{R}} \left(1 - e^{-\lambda|x|}\right) \Pi(dx)\right).$$

Letting $\lambda \downarrow 0$, monotone convergence on the left-hand side yields

$$q = \exp\left(-t \lim_{\lambda \downarrow 0} \int_{\mathbb{R}} \left(1 - e^{-\lambda|x|}\right) \Pi(dx)\right).$$

The limit in the exponent is zero. Indeed, on $\{|x| > 1\}$ we have $\Pi\left(\{|x| > 1\}\right) < \infty$, while on $\{|x| \le 1\}$, $1 - e^{-\lambda|x|} \le \lambda |x|$ and $\int_{|x| \le 1} |x| \, \Pi(dx) < \infty$. Thus the desired limit follows from the dominated convergence theorem. Consequently, $q = 1$, and hence

$$\sum_{s \le t} |\Delta_s| < \infty \quad \text{a.s.}$$

Since $t > 0$ was arbitrary, applying the result to all rational $t > 0$ and then using monotonicity gives the claim for every $t > 0$ almost surely. $\square$

For example, the measure

$$\Pi(dx) = |x|^{-\alpha-1} dx$$

satisfies the condition in (4.5) precisely when $\alpha \in (0, 1)$. If $\alpha \ge 1$, the integral in (4.5) diverges near zero. The next corollary shows that every Poisson random measure satisfying (4.5) generates a Lévy process of bounded variation.

Corollary 4.7. *If* Π *is a measure on* $\mathbb{R} \setminus \{0\}$ *such that*

$$\int_{\mathbb{R}} \min\{|x|, 1\}\Pi(dx) < \infty,$$

then the process

$$X = (X_t)_{t \geq 0} = \left(\sum_{s \leq t} \Delta_s \right)_{t \geq 0},$$

generated by the Poisson random measure with intensity $dt\, \Pi(dx)$*, has stationary and independent increments and paths of bounded variation. In particular,* X *is a Lévy process. Moreover, for every* $t > 0$*,*

$$\mathbb{P}\left(\Delta_t = 0\right) = 1.$$

Proof. Since

$$\int_{\mathbb{R}} \min\{|x|, 1\}\Pi(dx) < \infty,$$

Theorem 4.6 gives

$$\sum_{s \leq t} |\Delta_s| < \infty \qquad \text{a.s. for every } t > 0.$$

Thus X is well defined.

For $a > 0$, set

$$X_t^a = \sum_{s \leq t} \Delta_s 1_{|\Delta_s| > a}, \qquad t \geq 0.$$

Each X^a is a compound Poisson process, and therefore has stationary and independent increments. As $a \downarrow 0$, we have pathwise convergence

$$X_t^a \to X_t, \qquad t \geq 0,$$

and similarly for increments. Hence the stationary and independent increments pass to the limit. Indeed, if $\xi_n \perp\!\!\!\perp \eta_n$ and $(\xi_n, \eta_n) \xrightarrow{d} (\xi, \eta)$, then

$$\mathbb{E}\left[e^{i\theta\xi + i\zeta\eta}\right] = \lim_{n\to\infty} \mathbb{E}\left[e^{i\theta\xi_n}\right] \mathbb{E}\left[e^{i\zeta\eta_n}\right] = \mathbb{E}\left[e^{i\theta\xi}\right] \mathbb{E}\left[e^{i\zeta\eta}\right],$$

so $\xi \perp\!\!\!\perp \eta$. The same argument gives stationarity.

Next, for every $a > 0$ and $t > 0$,

$$\mathbb{P}\left(\Delta_t^a = 0\right) = 1.$$

Since

$$\{|\Delta_t| > 0\} = \bigcup_{r \in \mathbb{Q},\ r > 0} \{|\Delta_t| > r\},$$

we obtain $\mathbb{P}\left(\Delta_t = 0\right) = 1$.

Finally, write

$$X = X^+ - X^-,$$

where

$$X_t^+ = \sum_{s\le t} \Delta_s 1_{\Delta_s>0}, \qquad X_t^- = \sum_{s\le t} -\Delta_s 1_{\Delta_s<0}.$$

Since X^+ and X^- are increasing, they are almost surely right-continuous, and therefore so is X. In particular, X has bounded variation. $\square$

Now we can give the following definition.

Definition 4.1. *A Lévy process X is said to have **bounded variation** if*

$$X_t = \gamma t + \sum_{s\le t} \Delta_s,$$

where $\gamma \in \mathbb{R}$, and $(\Delta_s)_{s\ge 0}$ are the spatial coordinates of the points of a Poisson random measure with intensity $dt\,\Pi(dx)$, such that

$$\int_{\mathbb{R}} \min\{|x|, 1\}\Pi(dx) < \infty.$$

Being generated by a Poisson random measure, the jump part of X may be decomposed into independent Lévy processes by restricting the jumps to disjoint regions. For example, for $a > 0$,

$$X_t = \gamma t + X_t^a + \tilde{X}_t^a = \gamma t + \sum_{s\le t} \Delta_s 1_{|\Delta_s|>a} + \sum_{s\le t} \Delta_s 1_{|\Delta_s|\le a}.$$

Corollary 4.8. *If X is a Lévy process of bounded variation, then*

$$\begin{aligned} \mathbb{E}\left[e^{i\theta X_t}\right] &= e^{t\Psi(i\theta)}, \qquad \theta \in \mathbb{R}, \\ \Psi(i\theta) &= i\theta\gamma + \int_{\mathbb{R}} \left(e^{i\theta x} - 1\right)\Pi(dx). \end{aligned} \tag{4.6}$$

If $\mathbb{E}[|X_1|] < \infty$, then, for every $t > 0$,

$$\mathbb{E}[X_t] = \gamma t + t\int_{\mathbb{R}} x\Pi(dx). \tag{4.7}$$

If, in addition, $\mathbb{E}[X_1] = 0$ and $\mathbb{E}\left[X_1^2\right] < \infty$, then

$$Var(X_t) = \mathbb{E}\left[X_t^2\right] = t\int_{\mathbb{R}} x^2\Pi(dx). \tag{4.8}$$

Proof. Since

$$X_t = \gamma t + \lim_{a\to 0} X_t^a \quad \text{a.s.,} \qquad \Pi^a(dx) = 1_{|x|>a}\Pi(dx),$$

and X^a is a compound Poisson process with intensity $dt\,\Pi^a(dx)$, we get from (4.3) that

$$\begin{aligned} \mathbb{E}\left[e^{i\theta X_t}\right] &= e^{i\theta\gamma t} \lim_{a\to 0} \mathbb{E}\left[e^{i\theta X_t^a}\right] \\ &= e^{i\theta\gamma t} \lim_{a\to 0} e^{t\int_{|x|>a}(e^{i\theta x}-1)\Pi(dx)}. \end{aligned}$$

By dominated convergence,

$$\lim_{a\to 0}\int_{|x|>a}(e^{i\theta x}-1)\Pi(dx)=\int_{\mathbb{R}}(e^{i\theta x}-1)\Pi(dx).$$

Indeed, for $|x|\geq 1$, the integrand is bounded by 2, while for $|x|<1$,

$$\left|e^{i\theta x}-1\right|\leq|\theta|\,|x|\,.$$

Since

$$\int_{\mathbb{R}}\min\{|x|\,,1\}\Pi(dx)<\infty,$$

this proves (4.6).

Assuming $\mathbb{E}\left[|X_1|\right]<\infty$, differentiation at $\theta=0$ gives

$$i\mathbb{E}\left[X_t\right]=\frac{d}{d\theta}\mathbb{E}\left[e^{i\theta X_t}\right]\bigg|_{\theta=0}=i\gamma t+it\int_{\mathbb{R}}x\Pi(dx),$$

which proves (4.7). If $\mathbb{E}\left[X_1\right]=0$ and $\mathbb{E}\left[X_1^2\right]<\infty$, differentiating twice gives

$$\mathbb{E}\left[X_t^2\right]=t\int_{\mathbb{R}}x^2\Pi(dx),$$

and since $\mathbb{E}\left[X_t\right]=t\mathbb{E}\left[X_1\right]=0$, this is (4.8). □

Let us give several examples of Lévy processes of bounded variation.

(1) **Stable Lévy process .** Take

$$\Pi(dx)=|x|^{-\alpha-1}\,dx,\qquad\alpha\in(0,1),\qquad\gamma=0.$$

Then

$$\int_{\mathbb{R}}\min\{1,|x|\}\,|x|^{-\alpha-1}\,dx<\infty.$$

Furthermore,

$$\Psi(i\theta)=\int_{\mathbb{R}}\left(e^{i\theta x}-1\right)\Pi(dx)=|\theta|^{\alpha}\int_{\mathbb{R}}\left(e^{iy}-1\right)|y|^{-\alpha-1}\,dy=C\,|\theta|^{\alpha}\,,\tag{4.9}$$

where

$$C=-2\int_0^{\infty}(1-\cos y)\,y^{-\alpha-1}\,dy<0.$$

(2) **Gamma process.** Recall first that

$$\Gamma(\alpha)=\int_0^{\infty}x^{\alpha-1}e^{-x}\,dx,\qquad\alpha>0.$$

Let $X=(X_t)_{t\geq 0}$ be the Lévy process with Lévy measure

$$\Pi(dx)=\alpha x^{-1}e^{-\beta x}1_{x>0}\,dx,\qquad\alpha,\beta>0.\tag{4.10}$$

Since

$$\int_{\mathbb{R}}\min\{|x|\,,1\}\Pi(dx)=\alpha\int_0^{\infty}\min\{x,1\}x^{-1}e^{-\beta x}\,dx<\infty,$$

this is a Lévy process of bounded variation with $\gamma = 0$.

Its law at time $t > 0$ is

$$\mathbb{P}\left(X_t \in dx\right) = \frac{\beta^{\alpha t}}{\Gamma(\alpha t)} x^{\alpha t - 1} e^{-\beta x} 1_{x>0}\, dx. \tag{4.11}$$

Equivalently, X_t has Gamma distribution with parameters αt and β. The Lévy-Khintchine exponent is

$$\Psi(i\theta) = \int_0^\infty \left(e^{i\theta x} - 1\right) \alpha x^{-1} e^{-\beta x}\, dx = -\alpha \log\left(1 - \frac{i\theta}{\beta}\right), \qquad \theta \in \mathbb{R}. \tag{4.12}$$

Hence

$$\mathbb{E}\left[e^{i\theta X_t}\right] = e^{t\Psi(i\theta)} = \left(1 - \frac{i\theta}{\beta}\right)^{-\alpha t}, \qquad \theta \in \mathbb{R}.$$

(3) **Hyperexponential compound Poisson process.** Let

$$\Pi(dx) = \sum_{j=1}^n a_j e^{-a_j x} 1_{x>0}\, dx, \qquad a_j > 0, \qquad \gamma = 0.$$

Then $\Pi(\mathbb{R}) < \infty$, and hence the process is a compound Poisson process. Its Lévy-Khintchine exponent is

$$\begin{aligned}\Psi(i\theta) &= \int_0^\infty \left(e^{i\theta x} - 1\right) \sum_{j=1}^n a_j e^{-a_j x}\, dx \\ &= \sum_{j=1}^n \left(\frac{a_j}{a_j - i\theta} - 1\right) = \sum_{j=1}^n \frac{i\theta}{a_j - i\theta}.\end{aligned}$$

4.3. **Lévy processes with unbounded variation.** So far, we have described all Lévy processes that can be represented as a sum of jumps, a linear drift, and Brownian motion. We shall now see that other Lévy processes also exist; this was one of the surprising discoveries in the early development of the theory.

We begin with a lemma.

Lemma 4.9. *Let X be a Lévy process with $\mathbb{E}\left[|X_1|\right] < \infty$ and $\mathbb{E}\left[X_1^2\right] < \infty$. Then*

- $\mathbb{E}\left[X_t\right] = t\mathbb{E}\left[X_1\right]$*;*
- $Y_t = X_t - \mathbb{E}\left[X_t\right]$ *is a martingale and a Lévy process;*
- $Var\left(Y_t\right) = t\, Var(Y_1)$*.*

Moreover,

$$\mathbb{E}\left[\sup_{s \le t} |Y_s|^2\right] \le 4 Var(Y_t). \tag{4.13}$$

Proof. Set

$$m(t) := \mathbb{E}\left[X_t\right], \qquad v(t) := Var(X_t), \qquad t \geq 0.$$

We first prove the formula for the variance. Since $X_t - X_s \perp\!\!\!\perp X_s$, for $0 \leq s < t$, and $X_t - X_s \stackrel{d}{=} X_{t-s}$

$$v(t) = Var(X_t - X_s) + Var(X_s) = v(t-s) + v(s).$$

Thus v is additive on $\mathbb{R}^+$. Since $v \geq 0$, it is linear, and therefore

$$v(t) = t\, v(1).$$

Next, we prove that

$$m(t) = tm(1), \qquad t \geq 0.$$

For $q = m/n \in \mathbb{Q}_+$,

$$X_q = \sum_{k=1}^{m} \left(X_{k/n} - X_{(k-1)/n}\right),$$

where the summands are i.i.d. with the same law as $X_{1/n}$. Hence

$$m(q) = \frac{m}{n} m(1) = qm(1).$$

To extend this to all $t \geq 0$, note that $X_h \to 0$ in probability as $h \downarrow 0$. Since $\mathbb{E}\left[X_1^2\right] < \infty$, we have $v(h) = hv(1) \to 0$. Hence $X_h - m(h) \to 0$ in L^2, and therefore in probability. It follows that $m(h) \to 0$ as $h \downarrow 0$. Thus m is continuous at 0, and the additive identity gives $m(t) = tm(1)$ for all $t \geq 0$.

Now set

$$Y_t = X_t - \mathbb{E}\left[X_t\right] = X_t - t\mathbb{E}\left[X_1\right].$$

Since Y differs from X only by a deterministic linear drift, it is again a Lévy process. Also, for $0 \leq s < t$,

$$Y_t - Y_s = X_t - X_s - (t-s)\mathbb{E}\left[X_1\right].$$

Using $X_t - X_s \perp\!\!\!\perp \mathcal{F}_s$ and $\mathbb{E}\left[X_t - X_s\right] = (t-s)\mathbb{E}\left[X_1\right]$, we obtain

$$\mathbb{E}\left[Y_t \mid \mathcal{F}_s\right] = \mathbb{E}\left[Y_t - Y_s \mid \mathcal{F}_s\right] + Y_s = Y_s.$$

Thus Y is a martingale. Moreover,

$$Var(Y_t) = Var(X_t) = t\, Var(X_1) = t\, Var(Y_1).$$

Since Y is a square-integrable martingale, Doob's L^2 maximal inequality, see [11, Theorem 12.13], gives

$$\mathbb{E}\left[\sup_{s \leq t} |Y_s|^2\right] \leq 4\mathbb{E}\left[|Y_t|^2\right] = 4Var(Y_t),$$

which proves (4.13). $\square$

The question is whether one can construct Lévy processes beyond the bounded-variation case while preserving stationary and independent increments. Let $\mu(ds, dx)$ be a Poisson random measure on $\mathbb{R}^+ \times [-1, 1]$ with intensity $ds\,\Pi(dx)$, and for $0 < \epsilon < 1$ set

$$\Pi^\epsilon(dx) = \Pi(dx) 1_{|x|>\epsilon}.$$

Consider

$$X_t^\epsilon = \sum_{s\le t} \Delta_s^\epsilon - t \int_{\epsilon<|x|\le 1} x\Pi(dx).$$

Then

$$Y_t^\epsilon = \sum_{s\le t} \Delta_s^\epsilon$$

is a compound Poisson process with jumps satisfying $\epsilon < |x| \le 1$. Thus Y^ϵ is a Lévy process with finite first absolute moment, and by (4.7),

$$\mathbb{E}\left[Y_t^\epsilon\right] = t \int_{\epsilon<|x|\le 1} x\Pi(dx).$$

Consequently,

$$\mathbb{E}\left[X_t^\epsilon\right] = 0,$$

and X^ϵ is a Lévy process with characteristic function

$$\mathbb{E}\left[e^{i\zeta X_t^\epsilon}\right] = \mathbb{E}\left[e^{i\zeta Y_t^\epsilon}\right] e^{-i\zeta t \int_{\epsilon<|x|\le 1} x\Pi(dx)} = e^{t\int_{-1}^{1}\left(e^{i\zeta x}-1-i\zeta x\right)1_{|x|>\epsilon}\Pi(dx)}, \tag{4.14}$$

see (4.6).

The following lemma gives the condition on the Lévy measure under which the compensated processes X^ϵ converge.

Lemma 4.10. *The random variables X_t^ϵ converge in distribution for every $t > 0$ as $\epsilon \to 0$ if and only if*

$$\int_{-1}^{1} x^2 \Pi(dx) < \infty.$$

Moreover, in this case the finite-dimensional distributions

$$\left(X_{t_1}^\epsilon, X_{t_2}^\epsilon, \cdots, X_{t_n}^\epsilon\right)$$

converge for all $0 < t_1 < t_2 < \cdots < t_n$.

Proof. For $\zeta \in \mathbb{R}$, by (4.14),

$$\mathbb{E}\left[e^{i\zeta X_t^\epsilon}\right] = \exp\left(t \int_{-1}^{1} \left(e^{i\zeta x} - 1 - i\zeta x\right) 1_{|x|>\epsilon}\Pi(dx)\right). \tag{4.15}$$

Set

$$f_\zeta(x) = e^{i\zeta x} - 1 - i\zeta x.$$

Then

$$\Re f_\zeta(x) = \cos(\zeta x) - 1, \qquad \Im f_\zeta(x) = \sin(\zeta x) - \zeta x.$$

We first prove sufficiency. Assume that

$$\int_{-1}^{1} x^2 \Pi(dx) < \infty.$$

Using

$$1 - \cos x \le \frac{x^2}{2}, \qquad |\sin x - x| \le C x^2$$

for $|x| \le 1$, we get

$$|\Re f_\zeta(x)| \le \frac{\zeta^2 x^2}{2}, \qquad |\Im f_\zeta(x)| \le C_\zeta x^2, \qquad |x| \le 1.$$

Thus the real and imaginary parts are dominated by Π-integrable functions, and the dominated convergence theorem gives

$$\lim_{\epsilon \downarrow 0} \int_{-1}^{1} f_\zeta(x) 1_{|x|>\epsilon} \Pi(dx) = \int_{-1}^{1} f_\zeta(x) \Pi(dx).$$

Therefore

$$\lim_{\epsilon \downarrow 0} \mathbb{E}\left[e^{i\zeta X_t^\epsilon}\right] = \exp\left(t \int_{-1}^{1} \left(e^{i\zeta x} - 1 - i\zeta x\right) \Pi(dx)\right).$$

The limiting function is continuous at 0, so Lévy's continuity theorem gives convergence in distribution.

Conversely, assume that

$$\int_{-1}^{1} x^2 \Pi(dx) = \infty.$$

For $\zeta \neq 0$, Taylor's formula gives $c_\zeta > 0$ and $\delta_\zeta > 0$ such that

$$1 - \cos(\zeta x) \ge c_\zeta x^2, \qquad |x| \le \delta_\zeta.$$

Thus, as $\epsilon \downarrow 0$,

$$\int_{-1}^{1} (1 - \cos(\zeta x))\, 1_{|x|>\epsilon} \Pi(dx) \longrightarrow \infty.$$

Consequently, the real part of the exponent in (4.15) tends to $-\infty$, and hence

$$\lim_{\epsilon \downarrow 0} \mathbb{E}\left[e^{i\zeta X_t^\epsilon}\right] = 0, \qquad \zeta \neq 0.$$

The pointwise limit would therefore be 1 at $\zeta = 0$ and 0 for every $\zeta \neq 0$, which is not continuous at 0. By Lévy's continuity theorem, X_t^ϵ cannot converge in distribution. This proves necessity.

Finally, if $0 < t_1 < \cdots < t_n$, then the vector

$$\left(X_{t_1}^\epsilon, X_{t_2}^\epsilon - X_{t_1}^\epsilon, \cdots, X_{t_n}^\epsilon - X_{t_{n-1}}^\epsilon\right)$$

has independent components. Each component converges in distribution by the one-dimensional result. Hence the whole vector converges in distribution, and therefore so does

$$\left(X^{\epsilon}_{t_1}, X^{\epsilon}_{t_2}, \cdots, X^{\epsilon}_{t_n}\right).$$

$\square$

What is needed in order that $X^{\epsilon} = (X^{\epsilon}_t)_{t\geq 0}$ converge weakly to some process X? We have already proved convergence of the finite-dimensional distributions, and it remains to establish relative compactness (tightness).

Recall that tightness means that, for every $\eta > 0$, there exists a compact set K in the state space such that

$$\inf_{\epsilon} \mathbb{P}\left(X^{\epsilon} \in K\right) \geq 1 - \eta.$$

In our setting, the state space is typically the Skorokhod space $\mathcal{D}$. Thus tightness means that, uniformly in ϵ, the paths of X^{ϵ} remain with high probability in compact subsets of $\mathcal{D}$. By Prokhorov's theorem, tightness is equivalent to relative compactness of the corresponding laws. Hence, once tightness is proved, every sequence $\epsilon_n \downarrow 0$ has a weakly convergent subsequence, and the convergence of finite-dimensional distributions identifies the only possible limit.

For reference, the general theory of weak convergence and tightness on metric spaces is developed in Chapter 1 of Billingsley, while the corresponding criteria for the Skorokhod space $\mathcal{D}$ are treated in Chapter 3; see [4].

The next theorem characterizes all Lévy processes.

Theorem 4.11. *Let Π be a measure on $\mathbb{R}\setminus\{0\}$ such that*

$$\int_{\mathbb{R}} \min\{x^2, 1\}\Pi(dx) < \infty, \tag{4.16}$$

and let $a \in \mathbb{R}$, $\sigma^2 \in \mathbb{R}^+$. Then there exists a Lévy process $X = (X_t)_{t\in\mathbb{R}^+}$ such that, for $t \geq 0$,

$$\mathbb{E}\left[e^{i\zeta X_t}\right] = e^{t\Psi(i\zeta)}, \qquad \zeta \in \mathbb{R}, \tag{4.17}$$

where

$$\Psi\left(i\zeta\right) = ai\zeta - \frac{\sigma^2}{2}\zeta^2 + \int_{\mathbb{R}} \left(e^{i\zeta x} - 1 - i\zeta x 1_{|x|\leq 1}\right)\Pi(dx). \tag{4.18}$$

In probabilistic terms,

$$X_t = at + \sigma B_t + Y_t + Z_t, \qquad t \geq 0, \tag{4.19}$$

where B is a standard Brownian motion, Z is a compound Poisson process with intensity measure

$$dt\, 1_{|x|>1}\Pi(dx),$$

and Y is a martingale with zero mean, jumps bounded in absolute value by 1, and finite absolute moments of every order. The processes B, Y, and Z are mutually independent.

Remark 4.12. *The triplet $\left(a, \sigma^2, \Pi\right)$, consisting of $a \in \mathbb{R}$, the linear term, $\sigma^2 \in \mathbb{R}^+$, the Brownian term, and the measure Π, called the Lévy measure, is known as the Lévy triplet. Let us consider some examples:*

- *stable processes: $a = \sigma^2 = 0$ and*
$$\Pi(dx) = c_1 \frac{dx}{x^{1+\alpha}} 1_{x>0} + c_2 \frac{dx}{|x|^{1+\alpha}} 1_{x<0},$$
where $c_1, c_2 > 0$ and $\alpha \in (0, 2)$. Then
$$\int_{\mathbb{R}} \min\{x^2, 1\} \Pi(dx) < \infty.$$
When $c_1 = c_2$, we obtain symmetric stable processes with characteristic exponent
$$\Psi\left(i\zeta\right) = C \left|\zeta\right|^{\alpha},$$
where $C < 0$ is explicit. These processes arise as limits of normalised sums beyond the central limit theorem.
- *Gamma processes with measure*
$$\Pi(dx) = a x^{-1} e^{-bx} 1_{x>0}\, dx, \qquad a, b > 0.$$

Remark 4.13. *The Lévy–Itô decomposition can be seen from the proof of the theorem. In fact, many different decompositions are possible, corresponding to different truncations of the jumps.*

Proof. Take a standard Brownian motion B, and independently of it a compound Poisson process Z with intensity measure
$$dt\, 1_{|x|>1} \Pi(dx).$$
Then, from (4.4), with $\lambda F(dx) = 1_{|x|>1}\Pi(dx)$, the process
$$U_t = at + \sigma B_t + Z_t$$
satisfies

$$\mathbb{E}\left[e^{i\zeta U_t}\right] = e^{t\left(ai\zeta - \frac{\sigma^2}{2}\zeta^2 + \int_{|x|>1}\left(e^{i\zeta x} - 1\right)\Pi(dx)\right)}. \tag{4.20}$$

Next, for each $0 < \epsilon \leq 1$, take the compensated compound Poisson process X^{ϵ} defined in (4.14). We have already checked that
$$\mathbb{E}\left[X_t^{\epsilon}\right] = 0, \qquad t > 0,$$

and these processes have finite variance. Hence, by Lemma 4.9, X^ϵ is a martingale. Thus, for $0 < \gamma < \eta \le 1$, the difference

$$X^\gamma - X^\eta = (X_t^\gamma - X_t^\eta)_{t\ge 0}$$

is a square-integrable martingale with zero mean. By Doob's inequality, see (4.13), for every $t > 0$,

$$\mathbb{E}\left[\sup_{s\le t} |X_s^\eta - X_s^\gamma|^2\right] \le 4Var(X_t^\eta - X_t^\gamma).$$

On the other hand, $X^\gamma - X^\eta$ is the compensated compound Poisson process with intensity measure

$$dt\, 1_{\gamma<|x|\le\eta}\Pi(dx).$$

Therefore, by (4.8),

$$Var(X_t^\eta - X_t^\gamma) = t\int_{\gamma<|x|\le\eta} x^2\Pi(dx),$$

and hence

$$\mathbb{E}\left[\sup_{s\le t} |X_s^\eta - X_s^\gamma|^2\right] \le 4t\int_{\gamma<|x|\le\eta} x^2\Pi(dx) \le 4t\int_{0<|x|\le\eta} x^2\Pi(dx). \tag{4.21}$$

Let us summarise the properties obtained so far and introduce suitable notation. Fix $t > 0$. Let

$$Z^\eta := (X_s^\eta)_{s\le t}\,, \qquad \|Z\|_\infty = \sup_{s\le t} |Z_s|$$

for any stochastic process Z. We introduce the norm

$$\|Z\| := \left(\mathbb{E}\left[\|Z\|_\infty^2\right]\right)^{\frac{1}{2}} = \left(\mathbb{E}\left[\sup_{s\le t} |Z_s|^2\right]\right)^{\frac{1}{2}}.$$

Let

$$\mathbb{L} = \{Z : \|Z\| < \infty\}$$

be the space of stochastic processes on our underlying probability space whose norm is finite. We sketch the fact that $\mathbb{L}$ is a Banach space. If $(Z_n)_{n\ge1}$, with $Z_n \in \mathbb{L}$, is a Cauchy sequence in $\mathbb{L}$, then:

(1) since $\lim_{m,n\to\infty} \|Z_n - Z_m\| = 0$, we may choose $(n_k)_{k\ge1}$ so that

$$\|Z_{n_{k+1}} - Z_{n_k}\| \le \frac{1}{2^k};$$

(2) let

$$Y_k(s) = |Z_{n_1}(s)| + \sum_{j=2}^{k} \left|Z_{n_j}(s) - Z_{n_{j-1}}(s)\right|, \qquad s \le t;$$

then the triangle inequality gives

$$\|Y_k\| \le \|Z_{n_1}\| + \sum_{j=2}^{k} \|Z_{n_j} - Z_{n_{j-1}}\| \le \|Z_{n_1}\| + 2;$$

(3) $Y_k(s)$ is increasing in k for each $s \le t$, and hence there exists

$$Y(s) = \lim_{k\to\infty} Y_k(s) = |Z_{n_1}(s)| + \sum_{j=2}^{\infty} \left|Z_{n_j}(s) - Z_{n_{j-1}}(s)\right|, \qquad s \le t;$$

(4) by the monotone convergence theorem,

$$\begin{aligned}\|Y\|^2 &= \mathbb{E}\left[\left(\sup_{s\le t}|Y(s)|\right)^2\right] = \lim_{k\to\infty} \mathbb{E}\left[\left(\sup_{s\le t}|Y_k(s)|\right)^2\right] \\ &= \lim_{k\to\infty} \|Y_k\|^2 \le (\|Z_{n_1}\| + 2)^2 < \infty;\end{aligned}$$

(5) therefore $Y \in \mathbb{L}$, and since

$$Z_{n_k}(s) = Z_{n_1}(s) + \sum_{j=2}^{k} \left(Z_{n_j}(s) - Z_{n_{j-1}}(s)\right)$$

is absolutely dominated by $Y(s)$, it follows that $Z_{n_k}(s)$ converges pointwise to a limit T_s. Moreover,

$$\sup_{s\le t} \left|Z_{n_{k+m}}(s) - Z_{n_k}(s)\right| \le \sup_{s\le t} |Y_{k+m}(s) - Y_k(s)|,$$

and hence $Z_{n_k} \to T$ in $\mathbb{L}$;

(6) consequently, since for every $n \ge 1$,

$$\|Z_n - T\| \le \|Z_n - Z_{n_k}\| + \|Z_{n_k} - T\|,$$

we conclude that $Z_n \to T$ in $\mathbb{L}$;

(7) in particular,

$$\sup_{s\le t} |Z_n(s) - T_s| \xrightarrow{\mathbb{P}} 0,$$

and therefore there is a subsequence (η_i) such that

$$\sup_{s\le t} |Z^{\eta_i}(s) - T_s| \xrightarrow{\text{a.s.}} 0.$$

After establishing that $\mathbb{L}$ is a Banach space, we use (4.21), which may be rewritten as

$$\|Z^\eta - Z^\gamma\|^2 \le 4t \int_{0<|x|\le\eta} x^2 \Pi(dx). \tag{4.22}$$

Since $\int_{-1}^{1} x^2 \Pi(dx) < \infty$, we have

$$\lim_{\eta\to 0} \int_{0<|x|\le\eta} x^2 \Pi(dx) = 0.$$

Thus $(Z^{\eta})_{\eta>0}$ is a Cauchy family in $\mathbb{L}$, and therefore it has a limit, say T, realised as a uniform limit along a subsequence. Let us work with a subsequence Z^{η_i} such that almost surely

$$\lim_{i\to\infty}\sup_{s\le t}|T_s - Z^{\eta_i}(s)| = 0.$$

Clearly, for every $0 \le s < v \le t$,

$$Z^{\eta_i}(s) \to T_s \qquad \text{and} \qquad Z^{\eta_i}(v) - Z^{\eta_i}(s) \to T_v - T_s$$

in distribution. Since each Z^{η_i} has stationary and independent increments, we obtain

$$T_v - T_s \perp\!\!\!\perp T_s \qquad \text{and} \qquad T_v - T_s \overset{d}{=} T_{v-s}.$$

The same argument applies to finite collections of increments, so T has stationary and independent increments. Moreover, from (4.14),

$$\mathbb{E}\left[e^{i\zeta T_s}\right] = e^{s\int_{-1}^{1}\left(e^{i\zeta x}-1-i\zeta x\right)\Pi(dx)}, \qquad 0 \le s \le t. \tag{4.23}$$

Since $t > 0$ was arbitrary, these consistent limits define a process

$$T = (T_s)_{s\ge 0}$$

with stationary and independent increments. It remains to show that T has right-continuous paths.

Let us denote the jump sets of the processes by

$$A_i(\omega) = \{s \le t : |Z^{\eta_i}(s) - Z^{\eta_i}(s-)| > 0\}.$$

Then $A_i(\omega) \subseteq A_j(\omega)$ for $i < j$. Set

$$A(\omega) = \bigcup_i A_i(\omega).$$

We claim that $A(\omega)$ is exactly the set of jump times of $(T_s)_{s\le t}$. Indeed, by right-continuity of Z^{η_i},

$$\sup_{s\le t}|Z^{\eta_i}(s-) - Z^{\eta_j}(s-)| \le \sup_{s\le t}|Z^{\eta_i}(s) - Z^{\eta_j}(s)|\,.$$

Since for the chosen subsequence

$$\lim_{i,j\to\infty}\sup_{s\le t}|Z^{\eta_i}(s) - Z^{\eta_j}(s)| = 0,$$

it follows that $(Z^{\eta_i}(s-))_{i\ge 1}$ is a Cauchy sequence for every $s \le t$. Hence the left limits also converge. If $s \in A(\omega)$, then for all sufficiently large i,

$$|Z^{\eta_i}(s) - Z^{\eta_i}(s-)| > 0,$$

and so s is a jump time of T. Conversely, if $a \notin A(\omega)$ but $|T_a - T_{a-}| > u > 0$, then for every $i \geq 1$,

$$Z^{\eta_i}(a) = Z^{\eta_i}(a-),$$

which contradicts the uniform convergence of the subsequence to T, together with the convergence of the left limits. Therefore $A(\omega)$ is the set of jump times of T. Since T is the uniform limit on $[0,t]$ of right-continuous paths, T is right-continuous.

Finally, define

$$X_t = U_t + T_t.$$

Then X is a Lévy process, and combining (4.20) with (4.23) gives

$$\mathbb{E}\left[e^{i\zeta X_t}\right] = e^{t\Psi(i\zeta)},$$

with Ψ given by (4.18). □

Next we repeat some structural properties of Lévy processes.

5. General properties of Lévy processes

We describe some basic structural properties of Lévy processes.

5.1. Martingale property.

The following are basic martingale properties.

- a Lévy process is a martingale if and only if $\mathbb{E}[|X_1|] < \infty$ and $\mathbb{E}[X_1] = 0$; equivalently, if $\mathbb{E}[|X_1|] < \infty$, then

$$Y_t = X_t - t\mathbb{E}[X_1], \qquad t \geq 0,$$

  is a martingale;
- if $\mathbb{E}\left[X_1^2\right] < \infty$ and $\mathbb{E}[X_1] = 0$, then

$$Y_t = X_t^2 - t\mathbb{E}\left[X_1^2\right], \qquad t \geq 0,$$

  is a martingale;
- the process

$$M_t = e^{i\zeta X_t - t\Psi(i\zeta)}, \quad \zeta \in \mathbb{R},$$

  is a martingale, called the exponential martingale. Indeed, by the Lévy–Khintchine formula,

$$\mathbb{E}\left[e^{i\zeta(X_t - X_s)}\right] = e^{(t-s)\Psi(i\zeta)}, \qquad 0 \leq s < t.$$

Since $X_t - X_s \perp\!\!\!\perp \mathcal{F}_s$, we have

$$\begin{aligned}
\mathbb{E}\left[M_t \mid \mathcal{F}_s\right] &= \mathbb{E}\left[\left(e^{i\zeta X_t - t\Psi(i\zeta)}\right) \mid \mathcal{F}_s\right] \\
&= e^{-t\Psi(i\zeta)}\mathbb{E}\left[\left(e^{i\zeta(X_t - X_s)}e^{i\zeta X_s}\right) \mid \mathcal{F}_s\right] \\
&= e^{-t\Psi(i\zeta)}e^{i\zeta X_s}\mathbb{E}\left[e^{i\zeta(X_t - X_s)} \mid \mathcal{F}_s\right] \\
&= e^{-t\Psi(i\zeta)}e^{i\zeta X_s}\mathbb{E}\left[e^{i\zeta(X_t - X_s)}\right] \\
&= e^{-t\Psi(i\zeta)}e^{i\zeta X_s}e^{(t-s)\Psi(i\zeta)} \\
&= e^{i\zeta X_s - s\Psi(i\zeta)} = M_s.
\end{aligned}$$

5.2. **Strong Markov property.** We have already discussed the Markov property, but we record it here for completeness.

- a Lévy process is a Markov process. For $s \geq 0$, every bounded continuous function $f : \mathbb{R}^n \to \mathbb{R}$, and $t_1 < t_2 < \cdots < t_n$,

$$\mathbb{E}\left[f\left(X_{t_1+s}, \ldots, X_{t_n+s}\right)\Big|\mathcal{F}_s\right] = \int f\left(x_1 + X_s, \cdots, x_n + X_s\right)\mathbb{P}\left(X_{t_1} \in dx_1, \ldots, X_{t_n} \in dx_n\right)$$

  is X_s-measurable, and therefore almost surely

$$\mathbb{E}\left[f\left(X_{t_1+s}, \ldots, X_{t_n+s}\right)\Big|X_s\right] = \mathbb{E}\left[f\left(X_{t_1+s}, \ldots, X_{t_n+s}\right)\Big|\mathcal{F}_s\right].$$

- if τ is a stopping time, that is, $\{\tau \leq t\} \in \mathcal{F}_t$ for every $t \geq 0$, then on $\{\tau < \infty\}$, the process $\left(X_{\tau+t} - X_\tau\right)_{t\geq 0}$ is independent of $\left(X_t\right)_{t\leq\tau}$ and has the same law as X. This is the strong Markov property.

5.3. **Representation with Poisson random measure and compensation formula.** Recall that $X_t = at + \sigma B_t + Y_t + Z_t, t \geq 0$, is the Lévy–Itô decomposition of a Lévy process, X, see (4.19). The processes $Y = \left(Y_t\right)_{t\geq 0}, Z = \left(Z_t\right)_{t\geq 0}$ are associated to the Lévy measure Π and capture the compensated jumps of the process, which in turn stem from a Poisson random measure $\mu(dt, dx)$ with intensity $dt\Pi(dx)$. Therefore, from (2.1) and (2.2), which holds due to (4.16) we get that

$$Z_t = \int_0^t \int_{|x|>1} x\mu(ds, dx). \tag{5.1}$$

The approximation procedure, together with (4.18), show that

$$\begin{aligned}
&\lim_{\epsilon\to 0}\left(\sum_{s\leq t}\Delta_s^\epsilon - \int_0^t \int_{1\geq|x|>\epsilon} x\Pi(dx)ds\right) = \lim_{\epsilon\to 0}\int_0^t \int_{\epsilon<|x|\leq 1} x\tilde{\mu}(ds, dx) \\
&= Y_t = \int_0^t \int_{|x|\leq 1} x\tilde{\mu}(ds, dx), \text{ with } \tilde{\mu}(ds, dx) = \mu(ds, dx) - ds\Pi(dx)1_{|x|\leq 1},
\end{aligned} \tag{5.2}$$

where $\tilde{\mu}$ is the compensated Poisson random measure associated with μ.

Next, we record the compensation formula and some of its consequences; see [9, Chap. 4.3.2] for proofs. Assume that on $\{\Omega, \mathcal{F}, \mathbb{P}\}$, with filtration $(\mathcal{F}_t)_{t\geq 0}$, there is a non-negative stochastic process

$$\Phi = (\Phi(t,x)(\omega))_{t\geq 0,\ x\in\mathbb{R}}$$

such that, for every $t \geq 0$, the map $(x,\omega) \mapsto \Phi(t,x)(\omega)$ is $\mathcal{B}(\mathbb{R})\otimes\mathcal{F}_{t-}$-measurable; in particular, Φ is predictable. Then, for every $t \geq 0$,

$$\mathbb{E}\left[\int_0^t \int_{\mathbb{R}} \Phi(s,x)\mu(ds,dx)\right] = \mathbb{E}\left[\int_0^t \int_{\mathbb{R}} \Phi(s,x)\, ds\, \Pi(dx)\right]. \tag{5.3}$$

Equivalently, with (s, Δ_s) denoting the atoms of the Poisson random measure,

$$\mathbb{E}\left[\sum_{s\leq t} \Phi(s,\Delta_s)\right] = \int_0^t \int_{\mathbb{R}} \mathbb{E}\left[\Phi(s,x)\right]\, ds\, \Pi(dx), \tag{5.4}$$

see [3, Chap. 0]. One may freely take $\Phi(s,0) = 0, s \geq 0$.

6. Basic distributional properties of Lévy processes

Unlike Brownian motion, for which the distribution of B_t is known explicitly, for a general Lévy process the relevant information is encoded in Ψ, and explicit expressions for the law are usually not available. Here and throughout, we use the standard notation, for any $x \geq 0$,

$$\overline{\Pi}(x) = \Pi\left(\{|y| > x\}\right), \qquad \overline{\Pi}_+(x) = \Pi\left(\{y > x\}\right), \qquad \overline{\Pi}_-(x) = \Pi\left(\{-y > x\}\right), \tag{6.1}$$

for the two-sided, positive, and negative tails of the Lévy measure.

Before discussing the basic distributional properties, we collect some facts about the Lévy-Khintchine exponent (4.18). Recall the Lévy–Itô decomposition

$$X_t = at + B_t + Y_t + Z_t, \qquad t \geq 0,$$

where the processes B, Y, and Z are mutually independent. Then the Lévy-Khintchine exponent of

$$\tilde{X}_t = at + B_t + Y_t, \qquad t \geq 0,$$

is

$$\mathbb{E}\left[e^{i\zeta\tilde{X}_t}\right] = e^{t\left(ai\zeta - \frac{\sigma^2}{2}\zeta^2 + \int_{-1}^1\left(e^{i\zeta x} - 1 - i\zeta x\right)\Pi(dx)\right)} = e^{t\tilde{\Psi}(i\zeta)}, \qquad \zeta \in \mathbb{R},$$

see (4.18). Note that $\tilde{\Psi}$ extends to an entire function via

$$\tilde{\Psi}(z) = az - \frac{\sigma^2}{2}z^2 + \int_{-1}^1 \left(e^{zx} - 1 - zx\right)\Pi(dx), \qquad z \in \mathbb{C}.$$

This is because

$$\int_{-1}^1 |x|^n\, \Pi(dx) < \infty$$

for all $n \geq 2$. Thus, since $\tilde{\Psi}$ is entire,

$$\Psi(i\zeta) = \tilde{\Psi}(i\zeta) + \int_{|x|>1} \left(e^{i\zeta x} - 1\right) \Pi(dx). \tag{6.2}$$

Furthermore, we have the following useful asymptotic properties.

Proposition 6.1. *Let Ψ be a Lévy-Khintchine exponent. Then*

$$\lim_{|\theta|\to\infty} \frac{\Psi(i\theta)}{\theta^2} = -\frac{\sigma^2}{2}. \tag{6.3}$$

If $\sigma^2 = 0$ and $\int_{-1}^{1} |x|\Pi(dx) < \infty$, then

$$\lim_{|\theta|\to\infty} \frac{\Psi(i\theta)}{i\theta} = a - \int_{-1}^{1} x\Pi(dx). \tag{6.4}$$

Finally, Ψ is bounded on $i\mathbb{R}$ if and only if $\sigma^2 = 0$, $\Pi(\mathbb{R}) < \infty$, and $a = \int_{|x|\leq 1} x\Pi(dx)$.

Proof. The proofs of (6.3) and (6.4) rely on separating the real and imaginary parts of $e^{i\theta x} - 1$ using the dominated convergence theorem, and applying bounds of the form

$$|\sin x| \leq |x|, \qquad |1 - \cos(\theta x)| \leq 2\min\{x^2, 1\}.$$

For the last claim, see [3, Chap. I.1, Cor. 3]. ☐

We start with the discussion of the moments.

6.1. **Moments.** Since $\tilde{\Psi}$ is entire, $\tilde{X}_t$ possesses exponential moments of every complex order; see [8, Thm. 15.34].

(1) Exponential moments of X. From the decomposition above, $X = \tilde{X} + Z$, the existence of $\mathbb{E}\left[e^{aX_t}\right]$, $a \in \mathbb{R}$, is determined by the compound Poisson process Z, which captures the jumps of X whose absolute values exceed 1. Since $N_t \sim Poi(\overline{\Pi}(1)t)$, it follows from the exponential formula (2.3) that

$$\mathbb{E}\left[e^{i\theta Z_t}\right] = \mathbb{E}\left[e^{i\theta \sum_{j=1}^{N_t} \xi_j}\right] = e^{t\overline{\Pi}(1)\int_{|x|>1}\left(e^{i\theta x}-1\right)\mathbb{P}(\xi_1\in dx)}.$$

However, $\mathbb{P}(\xi_1 \in dx) = \Pi(dx)/\overline{\Pi}(1)$, and therefore the expression above extends from $i\theta \to a$ under the following equivalent conditions

$$\mathbb{E}\left[e^{aZ_t}\right] < \infty \iff \mathbb{E}\left[e^{a\xi_1}\right] < \infty \iff \int_{|x|>1} e^{ax}\Pi(dx) < \infty. \tag{6.5}$$

Thus the exponential moments of X exist if and only if the corresponding exponential moment of the Lévy measure is finite.

(2) Polynomial moments of X. Recall that moments are related to the derivatives of the characteristic function at zero, see the respective chapter in [8]. In particular, here, these are obtained by differentiation:

$$i^n \mathbb{E}\left[X_t^n\right] = \frac{d^n}{d\zeta^n} e^{t\Psi(i\zeta)}\bigg|_{\zeta=0}.$$

This requires Ψ to be at least n times differentiable at zero. For $n = 1$,

$$i\mathbb{E}\left[X_t\right] = t\Psi'(0) = it\left(a + \int_{|x|>1} x\Pi(dx)\right),$$

which requires $\mathbb{E}\left[|\xi_1|\right] < \infty$, or equivalently

$$\int_{|x|>1} |x|\Pi(dx) < \infty.$$

More generally, using again the Lévy–Itô decomposition and the fact that $\tilde{X}_t$ has exponential moments of all orders, we have, for any $t > 0$,

$$\mathbb{E}\left[\left|\tilde{X}_t\right|^k\right] < \infty, \qquad k \geq 0.$$

By Hölder's inequality, for $1 \leq k < n$,

$$\mathbb{E}\left[|Z_t|^k\right] \leq \left(\mathbb{E}\left[|Z_t|^n\right]\right)^{k/n}.$$

Hence, by the binomial formula

$$\mathbb{E}\left[|X_t|^n\right] < \infty \iff \mathbb{E}\left[|Z_t|^n\right] < \infty.$$

On the other hand, the inequality

$$\left(\sum_{j=1}^k |\xi_j|\right)^n \leq k^{n-1} \sum_{j=1}^k |\xi_j|^n$$

gives with the help that $N \perp\!\!\!\perp (\xi_j)_{j\geq 1}$

$$\mathbb{E}\left[|Z_t|^n\right] \leq \mathbb{E}\left[N_t^n\right] \mathbb{E}\left[|\xi_1|^n\right],$$
$$\mathbb{E}\left[|Z_t|^n\right] \geq \mathbb{E}\left[|\xi_1|^n \, 1_{N_t=1}\right] = \mathbb{P}\left(N_t = 1\right) \mathbb{E}\left[|\xi_1|^n\right].$$

Therefore, since N_t has finite integer moments,

$$\mathbb{E}\left[|X_t|^n\right] < \infty \iff \mathbb{E}\left[|\xi_1|^n\right] < \infty \iff \int_{|x|>1} |x|^n \Pi(dx) < \infty. \tag{6.6}$$

6.2. **Some basic facts about probability density functions.** If $f(\zeta) := |\Re(\Psi(i\zeta))| = -\Re(\Psi(i\zeta))$ is such that, for some $\epsilon > 0$,

$$\lim_{|\zeta|\to\infty} \frac{f(\zeta)}{|\zeta|^{\epsilon}} = \infty,$$

then X_t has a density g_t, which is infinitely differentiable. Indeed,

$$g_t(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ix\zeta} e^{t\Psi(i\zeta)}\, d\zeta, \tag{6.7}$$

and formal differentiation with respect to x gives

$$g_t^{(n)}(x) = \frac{(-i)^n}{2\pi} \int_{-\infty}^{\infty} \zeta^n e^{-ix\zeta} e^{t\Psi(i\zeta)}\, d\zeta.$$

Moreover,

$$\left|\zeta^n e^{-ix\zeta} e^{t\Psi(i\zeta)}\right| = |\zeta|^n e^{-tf(\zeta)} \le |\zeta|^n e^{-t|\zeta|^{\epsilon}},$$

and the integrability of the right-hand side guarantees the existence of the above integral.

A well-known sufficient condition for the infinite differentiability of g_t is Orey's condition; see [10, Thm. 28.3]. In terms of the Lévy measure, it may be written as

$$\liminf_{x\to 0} \frac{\int_{-x}^{x} y^2 \Pi(dy)}{x^2} > 0. \tag{6.8}$$

For example, Orey's condition holds if, for some $C > 0$, $\delta < 1$, and $\epsilon > 0$,

$$\Pi(dx) 1_{|x|<\delta} \ge C\, |x|^{-\epsilon}\, 1_{|x|<\delta}\, dx.$$

Stable processes satisfy this condition, as do many unbounded-variation Lévy processes.

It should be emphasised that (6.7) is important for the numerical approximation of the density of a Lévy process, since this density usually does not admit a closed-form expression.

6.3. **Time reversal duality.** For every $t > 0$, we have

$$(Y_s)_{s\le t} := \left(X_{(t-s)-} - X_t\right)_{s\le t} \stackrel{w}{=} (-X_s)_{s\le t}\,. \tag{6.9}$$

It is straightforward to verify the stationarity and independence of the increments of $(Y_s)_{s\le t}$, as well as the right-continuity of the paths; therefore, this is the restriction of a Lévy process to $[0,t]$. In addition, for any fixed $0 < s < t$,

$$X_{(t-s)-} - X_t \stackrel{a.s.}{=} X_{t-s} - X_t \stackrel{d}{=} -X_s,$$

which confirms the above identity.

Here and throughout, we write

$$S_t = \sup_{s\le t} X_s, \qquad I_t = \inf_{s\le t} X_s, \qquad t \ge 0,$$

for the running supremum and running infimum of the Lévy process X, respectively. Since (6.9) implies

$$\left(X_t - X_{(t-s)-}\right)_{s\leq t} \stackrel{w}{=} (X_s)_{s\leq t}\,,$$

it follows that

$$S_t \stackrel{d}{=} \sup_{s\leq t}\left(X_t - X_{(t-s)-}\right) = X_t - \inf_{s\leq t} X_{(t-s)-} = X_t - I_t.$$

Thus, we obtain the following link between the supremum and the infimum of a Lévy process.

Lemma 6.2. *For any Lévy process X and any $t > 0$, we have*

$$(X_t, S_t) \stackrel{d}{=} (X_t, X_t - I_t)\,. \tag{6.10}$$

This link will prove useful later on.

6.4. **Asymptotic behaviour of the paths.** If a Lévy process X satisfies $Var\,(X_1) < \infty$, then $(X_t - t\mathbb{E}\,[X_1])_{t\geq 0}$ contains the random walk $(X_n - n\mathbb{E}\,[X_1])_{n\geq 1}$, for which the law of the iterated logarithm holds. This extends readily to the whole process, yielding

$$\limsup_{t\to\infty} \frac{X_t - t\mathbb{E}\,[X_1]}{\sqrt{2t\log\log(t)}} = -\liminf_{t\to\infty} \frac{X_t - t\mathbb{E}\,[X_1]}{\sqrt{2t\log\log(t)}} = \sqrt{Var(X_1)}. \tag{6.11}$$

In addition, the central limit theorem holds:

$$\lim_{t\to\infty} \frac{X_t - t\mathbb{E}\,[X_1]}{\sqrt{tVar(X_1)}} \stackrel{d}{=} Z \sim N(0,1).$$

Assuming only that $\mathbb{E}\,[|X_1|] < \infty$, the strong law of large numbers holds:

$$\lim_{t\to\infty} \frac{X_t}{t} = \mathbb{E}\,[X_1] \text{ almost surely.}$$

Regardless of any further assumptions, Lévy processes obey the same trichotomy as random walks, in the sense that exactly one of the following holds almost surely:

(1) $\lim_{t\to\infty} X_t = \infty$;
(2) $\lim_{t\to\infty} X_t = -\infty$;
(3) $\limsup_{t\to\infty} X_t = \infty$ and $\liminf_{t\to\infty} X_t = -\infty$.

6.5. **Behaviour near 0.** One of the general results describing the behaviour near 0 is that for $x > 0$,

$$\lim_{t\to 0} \frac{\mathbb{P}\,(X_t > x)}{t} = \overline{\Pi}_+(x), \quad x > 0. \tag{6.12}$$

Since the jump process Δ stems from a Poisson random measure, the number of jumps exceeding x, namely

$$\#\{s \leq t : \Delta_s > x\},$$

is a Poisson random variable with parameter $t\overline{\Pi}_+(x)$. Therefore

$$\mathbb{P}\left(\exists\, s \le t : \Delta_s > x\right) = 1 - e^{-t\overline{\Pi}_+(x)} \sim t\overline{\Pi}_+(x).$$

Thus, (6.12) means that, over small time intervals, the event that X exceeds the level $x > 0$ is asymptotically caused by a single jump exceeding x. The proof of (6.12), however, is not trivial and requires either Fourier methods or more delicate probabilistic arguments.

If $\int_{-1}^{1} |x|\Pi(dx) < \infty$, then

$$\lim_{t\to 0} \frac{X_t}{t} \overset{\mathbb{P}}{=} a - \int_{-1}^{1} x\Pi(dx), \tag{6.13}$$

which follows from (6.4), since it implies

$$\lim_{t\to 0} \mathbb{E}\left[e^{\frac{i\theta}{t} X_t}\right] = \lim_{t\to 0} e^{t\Psi\left(\frac{i\theta}{t}\right)} = e^{i\theta\left(a - \int_{-1}^{1} x\Pi(dx)\right)}, \qquad \text{for all } \theta \in \mathbb{R}.$$

7. Subordinators

If X is a non-decreasing Lévy process, then its paths have bounded variation and it has no negative jumps. Consequently,

$$X_t = dt + \sum_{s\le t} \Delta_s, \qquad d \ge 0.$$

Since X is non-negative along its paths, the Lévy-Khintchine exponent takes the form

$$\Psi(z) = dz + \int_{x>0} \left(e^{zx} - 1\right)\Pi(dx), \qquad \Re(z) \le 0, \tag{7.1}$$

that is, it is analytic on $\mathbb{C}_-$. The fact that $d \ge 0$ follows from

$$\frac{X_t}{t} \overset{\mathbb{P}}{\to} d \ge 0,$$

see (6.13).

Application: If X is a Lévy process and T is an independent subordinator, then

$$Y_t = X_{T_t}, \qquad t \ge 0,$$

is again a Lévy process. A time change by T may be natural when the process evolves on the real time scale, but is observed intermittently. We speak of **anomalous diffusion** when

$$Y_t = X_{T_t^{-1}}, \qquad t \ge 0,$$

and such processes are used extensively in modelling. For example, this time change may describe the trapping of a freely moving particle, whose motion is governed by X, at spatial locations containing pores or traps.

Next, we slightly extend the notion of a subordinator by allowing killing. Let $\mathbf{e}_q \sim Exp(q)$, with the convention $\mathbf{e}_0 = \infty$, and assume that $\mathbf{e}_q \perp\!\!\!\perp X$. Define a process $Y = (Y_t)_{t\geq 0}$ by

$$Y_t = \begin{cases} X_t, & t < \mathbf{e}_q, \\ \infty, & t \geq \mathbf{e}_q. \end{cases}$$

Then Y is called a possibly killed Lévy process.

When dealing with subordinators, it is customary to work with

$$\phi(z) = -\log \mathbb{E}\left[e^{-zX_1}\right] = -\Psi(-z) = dz + \int_0^\infty \left(1 - e^{-zx}\right) \Pi(dx), \qquad \Re(z) \geq 0. \tag{7.2}$$

This function is called the Lévy-Khintchine exponent. We have the following immediate result.

Theorem 7.1. *Let Y be a possibly killed subordinator. Then*

$$\mathbb{E}\left[e^{-zY_t}\right] = e^{-t\phi(z)}, \qquad t \geq 0,\ \Re(z) \geq 0, \tag{7.3}$$

where

$$\phi(z) = dz + \int_0^\infty \left(1 - e^{-zx}\right) \Pi(dx) + q,$$

and d, Π are the characteristics of the underlying unkilled subordinator and q is the killing rate.

Remark 7.2. *Since the Lévy-Khintchine exponent of Y is directly related to that of X, we may treat all possibly killed subordinators as belonging to the same class. Moreover, in the form (7.2), the functions ϕ are precisely the Bernstein functions, that is, the functions whose first derivative is completely monotone and a Laplace transform of Radon measure. Indeed,*

$$\phi'(z) = d + \int_0^\infty e^{-zx} x\Pi(dx) = \int_0^\infty e^{-zx}\left(d\delta_0(dx) + x\Pi(dx)\right), \tag{7.4}$$

and therefore

$$(-1)^n \phi^{(n+1)}(x) \geq 0, \qquad x \in \mathbb{R}^+,\ n \geq 0.$$

Proof. For $\Re(z) > 0$, the claim follows directly from the definition of Y:

$$\begin{aligned} \mathbb{E}\left[e^{-zY_t}\right] &= \mathbb{E}\left[e^{-zX_t} 1_{t<\mathbf{e}_q}\right] \\ &= \mathbb{P}\left(t < \mathbf{e}_q\right) \mathbb{E}\left[e^{-zX_t}\right] \\ &= e^{-qt} \mathbb{E}\left[e^{-zX_t}\right]. \end{aligned}$$

The case $\Re(z) = 0$ is then obtained by a limiting argument. □

7.1. **Potential measure.** Here we define an important quantity, namely the potential measure. For any Borel set $A \subseteq \mathbb{R}^+$, we set

$$U(A) = \int_0^\infty \mathbb{P}\left(X_t \in A\right) \, dt = \mathbb{E}\left[\int_0^\infty 1_{X_t \in A} \, dt\right] \in [0, \infty] . \tag{7.5}$$

We note that U is σ-additive. Next, we show that it is finite on compact sets by computing its Laplace transform:

$$\widehat{U}(\lambda) = \int_0^\infty e^{-\lambda x} U(dx) = \int_0^\infty \mathbb{E}\left[e^{-\lambda X_t}\right] \, dt = \frac{1}{\phi(\lambda)} < \infty, \qquad \lambda > 0. \tag{7.6}$$

We also set

$$U(x) = U\left([0, x]\right), \qquad x \geq 0,$$

and call this the potential function. With

$$T_x = \inf\{t \geq 0 : X_t > x\}, \qquad x \geq 0, \tag{7.7}$$

it holds that

$$U(x) = \mathbb{E}\left[T_x\right],$$

and the strong Markov property easily yields

$$U(x + y) \leq U(x) + U(y), \qquad x, y > 0. \tag{7.8}$$

7.2. **Passage of level.** In this part we will give expressions for the law of $\left(X_{T_x}, X_{T_{x-}}\right), x > 0$, i.e. the positions after and prior to the passage. These give some tools to investigate the overshoot $(X_{T_x} - x)$ and the undershoot $(x - X_{T_{x-}})$.

Proposition 7.3. *Let X be a subordinator. Then, for any $x > 0$,*

$$\mathbb{P}\left(X_{T_x} \in dv, X_{T_{x-}} \in dy\right) = \Pi\left(dv - y\right) U(dy), \quad 0 \leq y \leq x < v. \tag{7.9}$$

Also, for any $x \geq 0$, $\mathbb{P}\left(X_{T_{x-}} < x = X_{T_x}\right) = 0$.

Proof. Note that for $y \leq x < v$,

$$\mathbb{P}\left(X_{T_{x-}} \leq y, \; X_{T_x} > v\right) = \mathbb{E}\left[\sum_{s \geq 0} 1_{X_{s-} \leq y} 1_{\Delta_s > v - X_{s-}}\right].$$

With

$$\Phi(s, u) = 1_{X_{s-} \leq y} 1_{u > v - X_{s-}},$$

formula (5.4) applies and yields

$$\mathbb{P}\left(X_{T_{x-}} \leq y, \; X_{T_x} > v\right) = \mathbb{E}\left[\int_0^\infty \int_{u \geq v - X_{s-}} 1_{X_{s-} \leq y} \Pi(du) \, ds\right].$$

Using that, for any fixed s, $X_s = X_{s-}$ almost surely, we obtain

$$\begin{aligned}\mathbb{P}\left(X_{T_x-} \le y,\ X_{T_x} > v\right) &= \int_0^y \int_v^\infty \mathbb{P}\left(X_{T_x-} \in dw, X_{T_x} \in d\rho\right)\\ &= \int_0^\infty \int_0^y \overline{\Pi}(v-w)\mathbb{P}\left(X_s \in dw\right)\, ds\\ &= \int_0^y \overline{\Pi}(v-w)\, U(dw)\\ &= \int_0^y \int_{(v,\infty)} \Pi(d\rho - w)\, U(dw),\end{aligned}$$

which proves (7.9).

Using the same argument, in the case $\Pi(\mathbb{R}) = \infty$ we obtain

$$\mathbb{P}\left(X_{T_x-} < x = X_{T_x}\right) = \int_0^x \Pi(\{x-y\})\, U(dy).$$

Let

$$\mathcal{A} = \{v : \Pi(\{v\}) > 0\}.$$

Then

$$\mathbb{P}\left(X_{T_x-} < x = X_{T_x}\right) = \sum_{v\in\mathcal{A}} \Pi(\{v\})\, U(\{x-v\}).$$

However, from the strong Markov property and $\Pi(\mathbb{R}) = \infty$, we see that, on $\{T_x < \infty\}$, with

$$T_{\{x\}} = \inf\{s > 0 : X_s = x\},$$

one has

$$\begin{aligned}U(\{x\}) = \mathbb{E}\left[\int_0^\infty 1_{X_t=x} dt\right] &= \mathbb{P}\left(T_{\{x\}} < \infty\right) \mathbb{E}\left[\int_{T_{\{x\}}}^\infty 1_{X_t=x} dt \middle| T_x < \infty\right]\\ &= \mathbb{P}\left(T_{\{x\}} < \infty\right) \mathbb{E}\left[\int_0^\infty 1_{X_{t+T_{\{x\}}}=x} dt \middle| T_x < \infty\right]\\ &= \mathbb{P}\left(T_{\{x\}} < \infty\right) \mathbb{E}\left[\int_0^\infty 1_{X_t=0} dt\right] = 0\end{aligned}$$

by the Strong Markov property and the fact that when $\Pi(\mathbb{R}) = \infty$ the process exits 0 instantaneously. Thus U is a diffuse measure. In the compound Poisson case, the event $\{X_{T_x-} < x = X_{T_x}\}$ cannot occur. □

We have the following result, due originally to Kesten; see also Andrew [1] for a much simpler proof.

Theorem 7.4. *Let X be a subordinator with $d = 0$. Then $\mathbb{P}(X_{T_x} > x) = 1$.*

When $d > 0$, it is known that $U(x)$ defines an absolutely continuous measure with bounded continuous density $u(x)$. In this case we have the following refinement.

Theorem 7.5. *Let X be a subordinator with $d > 0$. Then*

$$\mathbb{P}\left(X_{T_x} = x\right) = d\,u(x).$$

Recall that $\overline{\Pi}(x) = \Pi\left(\{|y| > x\}\right), x > 0$. These two theorems yield the following equation:

$$1 = \mathbb{P}\left(X_{T_x} \geq x\right) = d\,u(x) + \int_0^x \overline{\Pi}(x-w)U(dw). \tag{7.10}$$

In the case $d > 0$, this is a Volterra equation of the second kind, namely

$$1 = d\,u(x) + \int_0^x \overline{\Pi}(x-w)u(w)\,dw,$$

and

$$u(x) = \sum_{n\geq 0}(-1)^n d^{-n-1}(1 * \overline{\Pi})^{*n}(x), \qquad x > 0,$$

where

$$f * g(x) = \int_0^x f(x-y)g(y)\,dy,$$

see [6]. When $d = 0$ the equation cannot be solved even in semi-explicit form.

7.3. **Rates of growth.** Another important question regarding subordinators is their rate of growth both at zero and infinity. Especially relevant are the so-called laws of the iterated logarithm. We start with a sharper than (6.13) result at zero.

Theorem 7.6. *Let X be a subordinator. Then* $\lim_{t\to 0} X_t/t = d$ *almost surely.*

Proof. Without loss of generality, we may assume that X has no jumps larger than 1. The key point is that $(X_t/t)_{0<t\leq 1}$ is a reverse martingale. It is enough to work first on dyadic times in $(0, 1]$, namely

$$\left(\frac{X_{k2^{-n}}}{k2^{-n}} \,:\, 1 \leq k \leq 2^n,\ n \geq 1\right).$$

Fix $n \geq 1$, and write

$$S_k = X_{k2^{-n}}, \qquad 0 \leq k \leq 2^n.$$

Then

$$S_k = \sum_{j=1}^{k}\left(X_{j2^{-n}} - X_{(j-1)2^{-n}}\right) =: \sum_{j=1}^{k} Y_j,$$

where $(Y_j)_{1\leq j\leq 2^n}$ are independent and identically distributed, by stationarity and independence of the increments. Thus $(S_k)_{0\leq k\leq 2^n}$ is a random walk.

Since the variables $Y_1, \ldots, Y_{2^n}$ are exchangeable, for every $1 \leq j \leq 2^n$,

$$\mathbb{E}\left[Y_j \mid S_{2^n}\right] = \mathbb{E}\left[Y_1 \mid S_{2^n}\right].$$

Summing over $j = 1, \dots, k$, we obtain

$$\mathbb{E}\left[S_k \mid S_{2^n}\right] = \sum_{j=1}^{k} \mathbb{E}\left[Y_j \mid S_{2^n}\right] = k\,\frac{S_{2^n}}{2^n}.$$

Since $S_{2^n} = X_1$, this gives

$$\mathbb{E}\left[X_{k2^{-n}} \mid X_1\right] = k2^{-n} X_1,$$

or equivalently

$$\mathbb{E}\left[\frac{X_{k2^{-n}}}{k2^{-n}} \Big| X_1\right] = X_1.$$

Now let $0 < s < 1$, and choose $k(s,n)2^{-n} \uparrow s$. Then

$$\mathbb{E}\left[X_{k(s,n)2^{-n}} \mid X_1\right] = k(s,n)2^{-n} X_1.$$

Letting $n \to \infty$, and using monotone convergence together with $X_{s-} = X_s$ almost surely, we get

$$\mathbb{E}\left[X_s \mid X_1\right] = sX_1,$$

that is,

$$\mathbb{E}\left[\frac{X_s}{s} \Big| X_1\right] = \frac{X_1}{1}.$$

This proves the reverse martingale property.

Since $\mathbb{E}\left[X_t/t\right] = \mathbb{E}\left[X_1\right]$, the family $(X_t/t)_{0<t\leq 1}$ is L^1-bounded martingale. Hence, by the reverse martingale convergence theorem, see [7, Chap. 6, Thms. 6.18 and 6.27], X_t/t converges almost surely as $t \downarrow 0$. Since (6.13) gives convergence in distribution to the constant d, the almost sure limit must equal d. $\square$

We proceed with a law of the iterated logarithm for small times. Recall that a function $f : \mathbb{R}^+ \to \mathbb{R}^+$ is regularly varying at infinity/zero if for any $\lambda > 0$ and fixed $\alpha \in \mathbb{R}$

$$\lim_{x\to\infty/0} \frac{f(\lambda x)}{f(x)} = \lambda^{\alpha}. \tag{7.11}$$

For $\alpha = 0$ the function is called slowly varying. Any regularly varying function can be written as $f(x) = x^{\alpha} l(x)$, where l is slowly varying. We refer to the book [5] for more information on these functions.

In the sequel, the Bernstein function ϕ is assumed to be regularly varying at infinity with index $\alpha \in (0,1)$. Clearly, $\phi'(x) > 0$, and hence $\varphi = \phi^{-1}$ exists. It is well known that φ is regularly varying with index α^{-1}. We set

$$l_2(t) := \log\left|\log(t)\right|, \qquad t < e^{-1},$$

and

$$f(t) := \frac{l_2(t)}{\varphi\left(l_2(t)t^{-1}\right)}; \quad g(t) = \frac{f(t)}{t}.$$

Since φ is regularly varying of index $1/\alpha$ we get that, as $t \to 0$,

$$g(t) \sim \frac{t^{\frac{1}{\alpha}-1}}{l_2^{\frac{1}{\alpha}-1}(t)} \to 0.$$

The following theorem is proved in [3, Thm. 11].

Theorem 7.7. *Let X be a subordinator with Lévy-Khintchine exponent ϕ that is regularly varying at infinity with index $\alpha \in (0,1)$. Then, with f defined above,*

$$\liminf_{t \to 0} \frac{X_t}{f(t)} = \alpha(1-\alpha)^{\frac{1-\alpha}{\alpha}} \qquad \textit{almost surely.} \tag{7.12}$$

Proof. We sketch the main analytical idea behind the proof. The Esscher transform used in [3, Lem. 12] is the probabilistic analogue of the saddle-point method. Under mild assumptions, for $z = a + ib$ with $a > 0$,

$$\mathbb{E}\left[e^{-zX_t}\right] = \int_0^\infty e^{-zx}\mathbb{P}\left(X_t \in dx\right) = e^{-t\phi(z)} \implies$$
$$\mathbb{P}\left(X_t \le x\right) = \frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty} e^{zx}\frac{e^{-t\phi(z)}}{z}\,dz.$$

Hence

$$\mathbb{P}\left(X_t \le x\right) = \frac{e^{ax-t\phi(a)}}{2\pi}\int_{-\infty}^{\infty} e^{ibx}\frac{e^{-t(\phi(a+ib)-\phi(a))}}{a+ib}\,db. \tag{7.13}$$

Setting $x = cf(t)$, the leading term in $\ln \mathbb{P}\left(X_t \le cf(t)\right)$ is

$$acf(t) - t\phi(a).$$

Differentiating we get $a^*(t) = (\phi')^{-1}(cg(t))$ and since $g(t) \to 0$ then $a^*(t) \to \infty$. Also,

$$a^*cf(t) - t\phi(a^*) = t\left(cg(t)(\phi')^{-1}(cg(t))\right) - \phi\left((\phi')^{-1}\left((cg(t))\right)\right) = tH\left((\phi')^{-1}\left(cg(t)\right)\right)$$

with $H(x) = x\phi'(x) - \phi(x) = \int_0^x y\phi''(y)dy$ being decreasing since $\phi'' < 0$. For the sake of simplicity assume that

$$\phi(x) \sim x^\alpha.$$

Then $\phi'(x) \sim \alpha x^{\alpha-1}, \phi'(x) \sim \left(\frac{x}{\alpha}\right)^{\frac{1}{\alpha-1}}$ and $H(x) \sim -(1-\alpha)x^\alpha$. Thus,

$$tH\left((\phi')^{-1}\left(cg(t)\right)\right) \sim -t(1-\alpha)\left((\phi')^{-1}\left(cg(t)\right)\right)^\alpha \sim -t(1-\alpha)\left(\frac{cg(t)}{\alpha}\right)^{\frac{\alpha}{1-\alpha}}.$$

However, $g(t) \sim (l_2(t)/t)^{\frac{\alpha-1}{\alpha}}$ and hence

$$tH\left((\phi')^{-1}\left(cg(t)\right)\right) \sim -t(1-\alpha)\left(\frac{c}{\alpha}\right)^{\frac{\alpha}{1-\alpha}}\frac{l_2(t)}{t} \sim -(1-\alpha)\left(\frac{c}{\alpha}\right)^{\frac{\alpha}{1-\alpha}} l_2(t).$$

Therefore ignoring the integral term in (7.13) for any $r < 1$ and large n we get

$$\ln \mathbb{P}\left(X_{r^n} \le cf(r^n)\right) \sim \ln \left|\frac{1}{\ln r^n}\right|^{(1-\alpha)\left(\frac{c}{\alpha}\right)^{\frac{\alpha}{1-\alpha}}} (1+o(1)).$$

Reworking we obtain

$$\ln \mathbb{P}\left(X_{r^n} \le cf(r^n)\right) \sim \ln \left(\frac{1}{n}\right)^{(1-\alpha)\left(\frac{c}{\alpha}\right)^{\frac{\alpha}{1-\alpha}}}$$

and finally

$$\mathbb{P}\left(X_{r^n} \le cf(r^n)\right) \asymp \left(\frac{1}{n}\right)^{(1-\alpha)\left(\frac{c}{\alpha}\right)^{\frac{\alpha}{1-\alpha}}}.$$

The series

$$\sum_{n\ge 1} \mathbb{P}\left(X_{r^n} \le cf(r^n)\right) < \infty \iff (1-\alpha)\left(\frac{c}{\alpha}\right)^{\frac{\alpha}{1-\alpha}} > 1.$$

For all such $c > 0$ almost surely

$$\limsup_{n\to\infty} \frac{X_{r^n}}{f(r^n)} < c.$$

The rest is a interplay of varying r. The lower bound is obtained by considering

$$X_{\frac{r^n}{1-r}} - X_{\frac{r^{n+1}}{1-r}} \stackrel{d}{=} X_{r^n}$$

but are independent random variables and we can apply the second Borel-Cantelli.

More specific way would be to apply the saddle-point method chooses $a = a^*$ so that

$$\phi'(a^*) = c\frac{f(t)}{t} = c\frac{u(t)}{\varphi(u(t))}, \qquad u(t) = \frac{l_2(t)}{t}.$$

Since $u(t) \to \infty$ as $t \downarrow 0$ and $\phi(x)/x \to 0$ as $x \to \infty$, it follows that

$$\frac{u(t)}{\varphi(u(t))} \to 0,$$

and hence $a^*(t) \to \infty$.

Because ϕ is regularly varying,

$$\lim_{y\to\infty} \frac{y\phi'(y)}{\phi(y)} = \alpha,$$

so for large a the saddle-point equation is approximated by

$$\alpha\frac{\phi(a)}{a} \approx c\frac{u(t)}{\varphi(u(t))}.$$

Taking

$$a = \varphi(\beta u(t)),$$

we get

$$acf(t) - t\phi(a) = \varphi(\beta u(t))c\frac{l_2(t)}{\varphi(u(t))} - t\beta u(t) \stackrel{0}{\sim} l_2(t)\left(c\beta^{1/\alpha} - \beta\right).$$

This is minimised at

$$\beta = \left(\frac{\alpha}{c}\right)^{\frac{\alpha}{1-\alpha}},$$

and then, as $t \to 0$,

$$acf(t) - t\phi(a) \overset{0}{\sim} -l_2(t)(1-\alpha)\left(\frac{\alpha}{c}\right)^{\frac{\alpha}{1-\alpha}}.$$

Thus, roughly,

$$\ln \mathbb{P}\left(X_t \le cf(t)\right) \asymp -l_2(t)(1-\alpha)\left(\frac{\alpha}{c}\right)^{\frac{\alpha}{1-\alpha}}$$

and with $t = r^n, r < 1$,

$$\mathbb{P}\left(X_{r^n} \le cf(r^n)\right) \asymp n^{-\left(\frac{\alpha}{c}\right)^{\frac{\alpha}{1-\alpha}}}$$

and one can use Borel–Cantelli as in [3]. $\square$

8. Wiener–Hopf factorization

Recall that $S_t = \sup_{s \le t} X_s$ and $I_t = \inf_{s \le t} X_s$ are the running supremum and infimum of a Lévy process. The aim of this section is to derive the Wiener–Hopf factorization. In its simplest probabilistic form, let $\mathbf{e}_q \sim Exp(q)$, $q > 0$, with $\mathbf{e}_q \perp\!\!\!\perp X$. Then

$$X_{\mathbf{e}_q} = S_{\mathbf{e}_q} - \left(S_{\mathbf{e}_q} - X_{\mathbf{e}_q}\right), \qquad S_{\mathbf{e}_q} \perp\!\!\!\perp \left(S_{\mathbf{e}_q} - X_{\mathbf{e}_q}\right) \overset{d}{=} -I_{\mathbf{e}_q}. \tag{8.1}$$

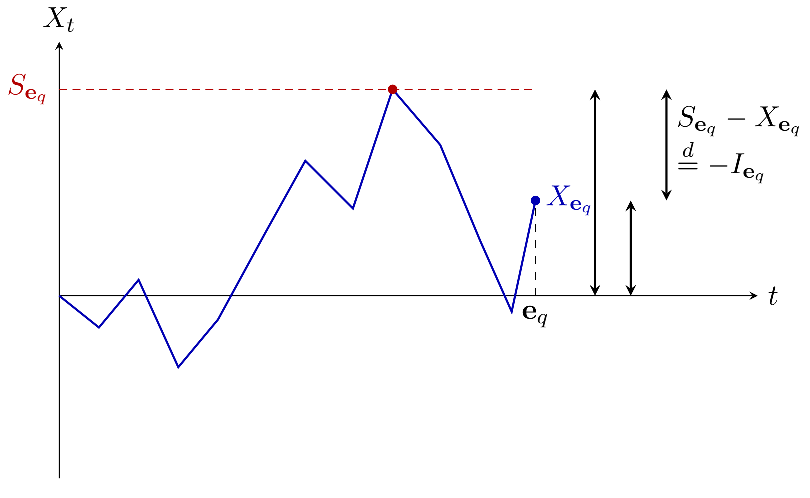

Figure 1. Wiener–Hopf factorization at the exponential time $\mathbf{e}_q$.

8.1. **Local time at the maximum.** Unlike a random walk, a Lévy process may attain infinitely many new maxima on each bounded time interval, so it is not convenient simply to track them one by one. For this purpose we consider the process reflected at its maximum,

$$R = (R_t)_{t\geq 0} = (S_t - X_t)_{t\geq 0}.$$

Clearly, R takes non-negative values and vanishes precisely at those times t for which $S_t = X_t$.

Proposition 8.1. *For any Lévy process X, the process R is a strong Markov process with respect to $(\mathcal{F}_t)_{t\geq 0}$, and for any $t \geq 0$,*

$$R_t \overset{d}{=} -I_t.$$

Proof. For $s, t \geq 0$,

$$\begin{aligned} R_{t+s} &= S_{t+s} - X_{t+s} \\ &= \max\{S_t, \sup_{t<v\leq t+s} X_v\} - X_{t+s} \\ &= \max\{S_t - X_t, \sup_{t<v\leq t+s} (X_v - X_t)\} - (X_{t+s} - X_t) \\ &= \max\{R_t, \tilde{S}_s\} - \tilde{X}_s, \end{aligned}$$

where $\tilde{X}_s = X_{t+s} - X_t$ and $\tilde{S}_s = \sup_{0\leq u\leq s} \tilde{X}_u$. By independence and stationarity of increments, $\tilde{X}$ is independent of $\mathcal{F}_t$ and has the same law as X. Hence R_{t+s} is a measurable function of R_t and an independent copy of X, which proves that R is a Markov process. The strong Markov property follows in the same way by replacing t with a stopping time and using the strong Markov property of X. The identity in law $R_t \overset{d}{=} -I_t$ follows from (6.10). □

Thus $\{t \geq 0 : R_t = 0\}$ is exactly the set of times when X is at its running maximum. We now introduce the notion of local time at the maximum.

Definition 8.1. *A local time at the maximum is an increasing process*

$$L = (L_t)_{t\geq 0},$$

adapted to $(\mathcal{F}_t)_{t\geq 0}$, such that:

(1) $L_0 = 0$;

(2) L increases only on the closure of the set

$$\{t \geq 0 : X_t = S_t\};$$

(3) for every stopping time τ such that $X_\tau = S_\tau$ on $\{\tau < \infty\}$, the shifted process

$$(L_{\tau+t} - L_\tau)_{t\geq 0}$$

is independent of $\mathcal{F}_\tau$ and has the same law as L.

Remark 8.2. *The definition above may be reformulated by requiring that, for every stopping time τ such that $R_\tau = 0$ on $\{\tau < \infty\}$, the shifted process*

$$\left(X^{(\tau)}, R^{(\tau)}, L_{\tau+\cdot} - L_\tau\right)$$

is independent of $\mathcal{F}_\tau$ and has the same law as (X, R, L). Moreover, if L is a local time at the maximum, then so is kL for any constant $k > 0$. Hence local time is defined only up to a multiplicative constant.

When it is necessary to distinguish between the clock L and the original parameter t, we shall refer to the latter as real time. By applying the same definition to $-X$, one may likewise introduce a local time at the minimum, which we denote by $\widehat{L}$.

Remark 8.3. *A continuous local time at the maximum need not always exist. In that case one may construct a right-continuous increasing process satisfying the same properties, and this is sufficient for the excursion analysis underlying the Wiener–Hopf factorization.*

To distinguish between the cases where a continuous local time exists and where it does not, we introduce the following notion.

Definition 8.2. *Let X be a Lévy process. We say that $\{0\}$ is regular for $(0, \infty)$ if, with*

$$T_{(0,\infty)} = \inf\{t \geq 0 : X_t > 0\},$$

we have $\mathbb{P}\left(T_{(0,\infty)} = 0\right) = 1$.

We note that $\mathcal{F}_0$ is trivial, see [3, Chap. I, Prop. 4], and therefore

$$\mathbb{P}\left(T_{(0,\infty)} = 0\right) \in \{0, 1\}.$$

Except in the compound Poisson case, where $\{0\}$ is not regular, the point $\{0\}$ is typically regular. In fact, this always holds when X has unbounded variation; see [9, Thm. 6.5]. We have the following theorem, see [9, Thm. 6.6].

Theorem 8.4. *Let X be a Lévy process, and let L denote a local time at the maximum.*

(i) *A continuous version of L exists if and only if $\{0\}$ is regular for $(0, \infty)$. When it exists, it is unique up to multiplication by a positive constant.*

(ii) *If $\{0\}$ is irregular for $(0, \infty)$, then one may take*

$$L_t = \sum_{j=0}^{n_t} \mathbf{e}_\lambda^{(j)}, \qquad t \geq 0, \tag{8.2}$$

where

$$n_t = \#\{0 < s \leq t : X_s = S_s\},$$

and $\left(\mathbf{e}_\lambda^{(j)}\right)_{j\geq 0}$ *are independent exponential random variables with parameter* $\lambda > 0$*, independent of* X*. This process satisfies the defining properties of local time.*

The next theorem shows when $\{s \geq 0 : R_s = 0\}$ has almost surely positive Lebesgue measure. We note in advance that regularity of $\{0\}$ for $(-\infty, 0)$ is defined analogously.

Theorem 8.5. *Assume that* X *is a Lévy process such that* $\{0\}$ *is regular for* $(0, \infty)$*, and let* L *be a continuous local time at the maximum. Then there exists a constant* $a \geq 0$ *such that, for every* $t \geq 0$*,*

$$\int_0^t 1_{X_s = S_s}\, ds = aL_t. \tag{8.3}$$

Moreover, $a > 0$ *if and only if* $\{0\}$ *is irregular for* $(-\infty, 0)$*.*

Proof. Set

$$A_t := \int_0^t 1_{X_s=S_s}\, ds = \int_0^t 1_{R_s=0}\, ds, \qquad t \geq 0.$$

Then $A = (A_t)_{t\geq 0}$ is continuous, increasing, adapted, and increases only on the set

$$\{t \geq 0 : X_t = S_t\} = \{t \geq 0 : R_t = 0\}.$$

Thus A has the same support of increase as the local time L.

Next, if τ is a stopping time such that $R_\tau = 0$ on $\{\tau < \infty\}$, then by the strong Markov property of R,

$$(R_{\tau+t})_{t\geq 0} \overset{w}{=} (R_t)_{t\geq 0} \qquad \text{and} \qquad (R_{\tau+t})_{t\geq 0} \perp\!\!\!\perp \mathcal{F}_\tau.$$

Therefore

$$A_{\tau+t} - A_\tau = \int_0^t 1_{R_{\tau+s}=0}\, ds$$

is independent of $\mathcal{F}_\tau$ and has the same law as A_t. Hence A satisfies the same regenerative property as L.

It follows that A is itself a local time at the maximum. Since continuous local time is unique up to multiplication by a positive constant, see Theorem 8.4, there exists $a \geq 0$ such that

$$A_t = aL_t, \qquad t \geq 0.$$

This proves (8.3). The final assertion concerning the sign of a is part of [9, Thm. 6.7]. $\square$

Since $\{0\}$ is regular for both $(0, \infty)$ and $(-\infty, 0)$ whenever X has unbounded variation, the constant a in (8.3) is equal to 0. Thus positive Lebesgue time can be be spent at the maximum only for certain classes of bounded-variation processes.

8.2. **The ascending ladder time and the ladder height processes.** We now define the ladder processes, which capture the times and levels at which a Lévy process reaches new maxima.

Definition 8.3. *Let $L = (L_t)_{t\geq 0}$ be a local time at the maximum.*

*The **ladder time process** is the right-continuous inverse*

$$L^{-1} = \left(L_t^{-1}\right)_{t\geq 0}, \qquad L_t^{-1} = \begin{cases} \inf\{s > 0 : L_s > t\}, & t < L_\infty, \\ \infty, & t \geq L_\infty. \end{cases}$$

*The **ladder height process** is the process*

$$H = (H_t)_{t\geq 0}, \qquad H_t = \begin{cases} X_{L_t^{-1}}, & t < L_\infty, \\ \infty, & t \geq L_\infty. \end{cases}$$

The main object of interest is the bivariate process

$$\left(L^{-1}, H\right) = \left(\left(L_t^{-1}, H_t\right)\right)_{t\geq 0}.$$

Similarly, by considering $\widehat{X} = -X$, we define the descending ladder time and ladder height processes of X as the corresponding ascending ladder processes of $\widehat{X}$, and denote them by

$$\left(\widehat{L}^{-1}, \widehat{H}\right).$$

A simple illustration is supplied in the next figure.

Next, we show that L_t^{-1} and L_{t-}^{-1} are stopping times. Recall that the underlying filtration is right-continuous; see Remark 3.2.

Lemma 8.6. *For any $t > 0$, both L_t^{-1} and L_{t-}^{-1} are stopping times with respect to $(\mathcal{F}_s)_{s\geq 0}$.*

Proof. Fix $t > 0$ and $s > 0$. Since $L_t^{-1} = \inf\{u > 0 : L_u > t\}$, we have

$$\{L_t^{-1} \leq s\} = \bigcap_{n\geq 1} \{L_{s+\frac{1}{n}} > t\} \in \bigcap_{n\geq 1} \mathcal{F}_{s+\frac{1}{n}} = \mathcal{F}_s.$$

Similarly,

$$\{L_{t-}^{-1} \leq s\} = \bigcap_{n\geq 1} \{L_{t-\frac{1}{n}}^{-1} \leq s\} \in \mathcal{F}_s,$$

and the proof is complete. $\square$

The first main result shows that $\left(L^{-1}, H\right)$ is a possibly killed bivariate subordinator. Of course, the same holds for $\left(\widehat{L}^{-1}, \widehat{H}\right)$. Recall briefly that a bivariate subordinator is described by its Lévy-Khintchine exponent in much the same way as in the one-dimensional case. Indeed, if

$$X = \left(X^{(1)}, X^{(2)}\right)$$

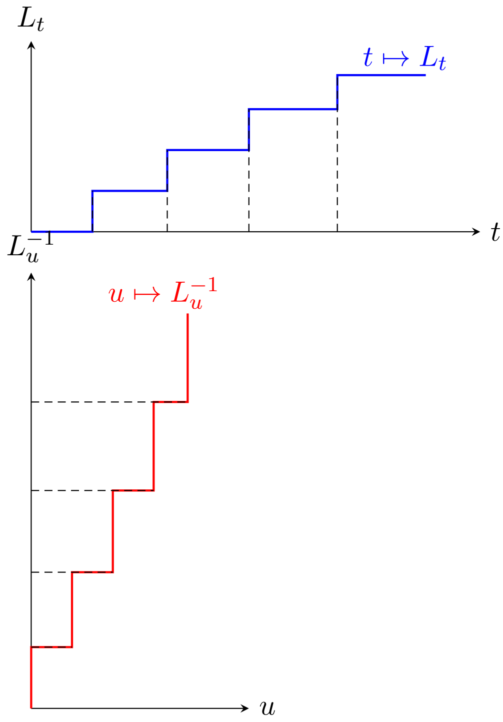

FIGURE 2. Relation between the local time L and L^{-1}.

is a bivariate subordinator, then

$$\mathbb{E}\left[e^{-z_1 X_t^{(1)} - z_2 X_t^{(2)}}\right] = e^{-t\phi(z_1,z_2)}, \qquad \Re(z_1) \geq 0,\ \Re(z_2) \geq 0,$$

(8.4)

$$\phi(z_1,z_2) = \phi(0,0) + d_1 z_1 + d_2 z_2 + \int_0^\infty \int_0^\infty (1 - e^{-z_1 x_1 - z_2 x_2}) \mu(dx_1, dx_2),$$

where $\phi(0,0) \geq 0$ is the killing rate, $d_1, d_2 \geq 0$ are the drift coefficients, and for the Lévy measure we have the condition

$$\int_0^\infty \int_0^\infty \min\{1, \sqrt{t^2 + x^2}\}\, \mu(dt, dx) < \infty.$$

Thus there is no substantial difference from the one-dimensional case.

Theorem 8.7. *Let X be a Lévy process . If*

$$\limsup_{t\to\infty} X_t = \infty \qquad \textit{almost surely},$$

then (L^{-1}, H) is an unkilled bivariate subordinator. Otherwise, (L^{-1}, H) is a killed bivariate subordinator with bivariate Lévy-Khintchine exponent κ, and

$$L_\infty = \lim_{t\to\infty} L_t \overset{d}{=} \mathbf{e}_q, \qquad q = \kappa(0,0).$$

Proof. We discuss only the case in which $\{0\}$ is regular for $(0,\infty)$, so that L is continuous. The other case is treated in [9, Thm. 6.9].

Fix $t>0$, and work on the event $\{L_\infty > t\}$. By continuity of L and the definition of L^{-1}, we have

$$L_{L_t^{-1}} = t.$$

Moreover,

$$R_{L_t^{-1}} = 0.$$

Indeed, since L_t^{-1} is a stopping time, if $R_{L_t^{-1}} > 0$, then by right-continuity we would have

$$X_{L_t^{-1}+h} < S_{L_t^{-1}}$$

for all sufficiently small $h>0$. Hence L would remain constant immediately after L_t^{-1}, which contradicts the definition of L_t^{-1}.

Now consider the shifted process

$$\tilde{X} = X^{(L_t^{-1})} = \left(X_{L_t^{-1}+s} - X_{L_t^{-1}}\right)_{s\geq 0}.$$

Its local time at the maximum is

$$\tilde{L} = L^{(L_t^{-1})} = \left(L_{L_t^{-1}+s} - t\right)_{s\geq 0}.$$

By the strong Markov property and item (3) of Definition 8.1, we have

$$\tilde{L} \overset{w}{=} L, \qquad \tilde{L} \perp\!\!\!\perp_{\{L_\infty > t\}} \mathcal{F}_{L_t^{-1}}.$$

Clearly,

$$\tilde{L}^{-1} = L_{t+\cdot}^{-1} - L_t^{-1}, \qquad \tilde{H} = \tilde{X}_{\tilde{L}^{-1}} = H_{t+\cdot} - H_t.$$

Hence

$$(\tilde{L}^{-1}, \tilde{H}) \perp\!\!\!\perp \mathcal{F}_{L_t^{-1}} \qquad \text{and} \qquad (\tilde{L}^{-1}, \tilde{H}) \overset{w}{=} (L^{-1}, H).$$

Therefore (L^{-1}, H) has stationary independent increments and is a possibly killed bivariate subordinator.

For $\lambda_1 = \lambda_2 = 0$, the same argument gives

$$\mathbb{P}\left(L_\infty > t+s\right) = \mathbb{P}\left(L_\infty > t\right)\mathbb{P}\left(L_\infty > s\right),$$

and hence L_∞ is exponentially distributed with parameter $\kappa(0,0)$. $\square$

Since L^{-1} is a possibly killed subordinator, the excursions of X away from its running maximum are naturally identified with the intervals

$$(L_{t-}^{-1}, L_t^{-1}), \qquad L_t^{-1} > L_{t-}^{-1}, \qquad t\geq 0.$$

Also, with

$$\Delta L^{-1} = \left(\Delta L_t^{-1}\right)_{t\geq 0} = \left(L_t^{-1} - L_{t-}^{-1}\right)_{t\geq 0},$$

we have on $\{L_\infty > t\}$,

$$L_t^{-1} - \sum_{s\leq t} \Delta L_s^{-1} = at.$$

On the other hand,

$$\left[0, L_t^{-1}\right] \setminus \bigcup_{v\leq t:\, L_v^{-1} > L_{v-}^{-1}} (L_{v-}^{-1}, L_v^{-1})$$

consists of times at which $R_s = 0$, up to endpoints of excursion intervals, and these endpoints have Lebesgue measure zero. Hence

$$at = \int_0^{L_t^{-1}} 1_{R_s=0}\, ds.$$

Therefore,

$$L_t^{-1} = \int_0^{L_t^{-1}} 1_{R_s=0}\, ds + \sum_{s\leq t} \Delta L_s^{-1}. \tag{8.5}$$

The same can be done for the local time of $X_t - I_t$ to get $\left(\hat{L}^{-1}, \hat{H}\right)$ the bivariate descending ladder time and ladder height process which of course is possibly killed subordinator too.

We now introduce the individual excursions.

Definition 8.4. *For each local-time level $t > 0$, we define the corresponding excursion from the maximum by*

$$\epsilon_t = \begin{cases} \left(X_{L_{t-}^{-1}+s} - X_{L_{t-}^{-1}}\right)_{0<s\leq L_t^{-1} - L_{t-}^{-1}}, & \text{if } L_{t-}^{-1} < L_t^{-1}, \\ \partial, & \text{if } L_{t-}^{-1} = L_t^{-1}, \end{cases}$$

where we adopt the convention $L_{0-}^{-1} = 0$, and ∂ is a cemetery state.

When $\epsilon_t \neq \partial$, we call ϵ_t the excursion from the maximum associated with local time t. Thus, for fixed $t > 0$, ϵ_t is itself a path, given by

$$\epsilon_t(s) = X_{L_{t-}^{-1}+s} - X_{L_{t-}^{-1}}, \qquad 0 < s \leq L_t^{-1} - L_{t-}^{-1}.$$

Definition 8.5. *Let $\mathcal{E}$ denote the space of excursions from the maximum. An element $\epsilon \in \mathcal{E}$ is a càdlàg path with lifetime $\zeta = \zeta(\epsilon) \in (0, \infty]$ such that*

$$\epsilon : (0, \zeta) \to (-\infty, 0),$$

and, when $\zeta < \infty$, the endpoint satisfies

$$\epsilon(\zeta) \in [0, \infty).$$

The quantity

$$h(\epsilon) := \epsilon(\zeta)$$

is called the terminal value of the excursion, and

$$\overline{\epsilon} := - \inf_{s \in (0,\zeta)} \epsilon(s)$$

is called its depth.

We note that

$$\mathcal{E} = \bigcup_{a>0} \mathcal{E}^a, \qquad \mathcal{E}^a = \{\epsilon \in \mathcal{E} : \zeta(\epsilon) > a\}.$$

On each $\mathcal{E}^a$ we define a probability measure by

$$\mathbb{P}_a(A) = \mathbb{P}\left(\epsilon \in A \mid \zeta(\epsilon) > a\right),$$

for any measurable $A \subseteq \mathcal{E}^a$, where measurability is understood with respect to the sigma-algebra induced by the Skorokhod topology. We then introduce the finite measure

$$n^a(A) := \overline{\Pi}_{L^{-1}}(a)\, \mathbb{P}_a(A), \tag{8.6}$$

where

$$\overline{\Pi}_{L^{-1}}(a) = \Pi_{L^{-1}}((a, \infty))$$

is the tail of the Lévy measure of L^{-1}. Since L^{-1} is a possibly killed subordinator, for $0 < b < a$ we have

$$\mathbb{P}\left(\zeta(\epsilon) > a \mid \zeta(\epsilon) > b\right) = \frac{\overline{\Pi}_{L^{-1}}(a)}{\overline{\Pi}_{L^{-1}}(b)}.$$

Hence the family $\left(n^a\right)_{a>0}$ is consistent, in the sense that on $\mathcal{E}^a \subseteq \mathcal{E}^b$,

$$n^b_{|\mathcal{E}^a} = n^a.$$

Therefore, there exists a unique σ-finite measure n on $\mathcal{E}$ such that, for every $a > 0$,

$$n_{|\mathcal{E}^a} = n^a.$$

Note that $\mathcal{E}^\infty$ may carry positive n-mass if and only if L^{-1} is killed, that is, if $L_\infty < \infty$. In that case,

$$n(\mathcal{E}^\infty) = \overline{\Pi}_{L^{-1}}(\infty) = \kappa(0,0) > 0,$$

see Theorem 8.7.

Theorem 8.8. *Let X be a Lévy process , and let n be the corresponding excursion measure on $\mathcal{E}$. Then the family of excursions*

$$e = (\epsilon_t)_{0 \leq t < L_\infty} ,$$

see Definition 8.4, forms a Poisson point process on $\mathcal{E}$ with characteristic measure n, stopped at the first point belonging to $\mathcal{E}^\infty$.

Proof. Without loss of generality, we work in the case $L_\infty = \infty$ almost surely. Fix $a > 0$, and consider

$$N_t^a = \#\{s \leq t : \zeta(\epsilon_s) > a\}, \qquad t \geq 0.$$

Since the jumps of L^{-1} correspond exactly to excursion intervals, we have

$$N_t^a = \#\{s \leq t : \Delta L_s^{-1} > a\}.$$

As L^{-1} is a subordinator with Lévy measure $\Pi_{L^{-1}}$, it follows that

$$N_t^a \sim Poi\left(t\,\overline{\Pi}_{L^{-1}}(a)\right),$$

and $(N_t^a)_{t\geq 0}$ is a Poisson process.

Now fix a measurable set $A \subseteq \mathcal{E}^a$, and define

$$N_t^{a,A} = \#\{s \leq t : \zeta(\epsilon_s) > a,\ \epsilon_s \in A\}, \qquad t \geq 0.$$

Conditional on the event $\{\zeta(\epsilon_s) > a\}$, the excursion ϵ_s has law $\mathbb{P}_a$. Hence $N^{a,A}$ is obtained from N^a by thinning with retention probability

$$\mathbb{P}_a(A) = \mathbb{P}\left(\epsilon \in A \mid \zeta(\epsilon) > a\right).$$

Therefore $N^{a,A}$ is a Poisson process with rate

$$\overline{\Pi}_{L^{-1}}(a)\mathbb{P}_a(A) = n(A),$$

that is,

$$N_t^{a,A} \sim Poi\left(t\,n(A)\right).$$

Since L^{-1} has independent and stationary increments, the same is true for the counts of its jumps in disjoint local-time intervals. Therefore $N^{a,A}$ has independent and stationary increments. Moreover, if $A, B \subseteq \mathcal{E}^a$ are disjoint, then $N^{a,A}$ and $N^{a,B}$ never jump simultaneously, and their increments over disjoint local-time intervals are independent. Hence the family of counts

$$\left(N_t^{a,A}\right)_{t\geq 0,\ A\subseteq\mathcal{E}^a}$$

defines a Poisson point process on $\mathcal{E}^a$ with intensity $dt\,n(d\epsilon)$.

Since $a > 0$ is arbitrary and $\mathcal{E} = \bigcup_{a>0} \mathcal{E}^a$, the family $(\epsilon_t)_{t\geq 0}$ is a Poisson point process on $\mathcal{E}$ with characteristic measure n. In the general case $L_\infty < \infty$, the process is stopped at the first point in $\mathcal{E}^\infty$. $\square$

8.3. **Wiener–Hopf factorization.** In this part we discuss the Wiener–Hopf factorization, assuming that the Lévy process is not a CPP. By Theorem 8.7, the processes

$$\left(L^{-1}, H\right) \qquad \text{and} \qquad \left(\widehat{L}^{-1}, \widehat{H}\right)$$

are bivariate subordinators. We denote their bivariate Laplace exponents by κ and $\widehat{\kappa}$, respectively.

For $t > 0$, set

$$\bar{G}_t = \sup\{s < t : S_s = X_s\}, \qquad \underline{G}_t = \sup\{s < t : I_s = X_s\}, \tag{8.7}$$

the times of the last supremum and infimum before t, respectively. The following proposition holds.

Proposition 8.9. *Let X be a Lévy process that is not a CPP. Then, for any fixed $s < t$, almost surely on the event*

$$\{\exists u \in (s,t) : S_u = X_u\},$$

we have

$$S_s < S_t.$$

The analogous statement holds for the infimum process.

The next lemma collects some preliminary results.

Lemma 8.10. *Let X be a Lévy process that is not a CPP, and let $\mathbf{e}_q$, $q > 0$, be an exponential random variable with parameter q, independent of X. Then*

$$\left(\mathbf{e}_q, X_{\mathbf{e}_q}\right)$$

is infinitely divisible with Lévy measure

$$\mu(dt, dx) = t^{-1} e^{-qt} \mathbb{P}\left(X_t \in dx\right)\, dt.$$

Furthermore, the pairs

$$\left(\bar{G}_{\mathbf{e}_q}, S_{\mathbf{e}_q}\right) \qquad \text{and} \qquad \left(\underline{G}_{\mathbf{e}_q}, -I_{\mathbf{e}_q}\right)$$

are infinitely divisible and satisfy

$$\mathbb{E}\left[e^{-\alpha \bar{G}_{\mathbf{e}_q} - \beta S_{\mathbf{e}_q}}\right] = \frac{\kappa(q,0)}{\kappa(q+\alpha, \beta)}, \qquad \Re(\alpha) \geq 0,\ \Re(\beta) \geq 0. \tag{8.8}$$

The analogous identity holds for

$$\left(\underline{G}_{\mathbf{e}_q}, -I_{\mathbf{e}_q}\right)$$

with κ replaced by $\widehat{\kappa}$. Finally,

$$\left(\bar{G}_{\mathbf{e}_q}, S_{\mathbf{e}_q}\right) \perp\!\!\!\perp \left(\mathbf{e}_q - \bar{G}_{\mathbf{e}_q}, S_{\mathbf{e}_q} - X_{\mathbf{e}_q}\right) \stackrel{d}{=} \left(\underline{G}_{\mathbf{e}_q}, -I_{\mathbf{e}_q}\right).$$

Proof. We prove only the first claim; the other two can be found in [3, Chap. VI] and require the compensation formula and excursion theory. For $\alpha \geq 0$ and $\beta \in \mathbb{R}$, we have

$$\begin{aligned}
\mathbb{E}\left[e^{-\alpha \mathbf{e}_q + i\beta X_{\mathbf{e}_q}}\right] &= q\int_0^\infty e^{-qt-\alpha t+t\Psi(i\beta)}\, dt = \frac{q}{q+\alpha-\Psi(i\beta)} \\
&= \exp\left(\int_0^\infty \left(e^{-(q+\alpha-\Psi(i\beta))t} - e^{-qt}\right)\frac{dt}{t}\right) \\
&= \exp\left(\int_0^\infty \int_{\mathbb{R}} \left(e^{-\alpha t+i\beta x} - 1\right)\mathbb{P}\left(X_t \in dx\right)\, e^{-qt}\frac{dt}{t}\right),
\end{aligned}$$

where we have used the Frullani integral identity

$$\log\left(\frac{q}{q+\alpha-\Psi(i\beta)}\right) = \int_0^\infty \left(e^{-(q+\alpha-\Psi(i\beta))t} - e^{-qt}\right)\frac{dt}{t}.$$

This is precisely the Lévy–Khintchine representation of the infinitely divisible random vector

$$\left(\mathbf{e}_q, X_{\mathbf{e}_q}\right)$$

with Lévy measure

$$\mu(dt, dx) = t^{-1}e^{-qt}\mathbb{P}\left(X_t \in dx\right)\, dt.$$

$\square$

We have the following main theorem.

Theorem 8.11. *Let X be a Lévy process that is not a CPP and let $\mathbf{e}_q \sim Exp(q)$, $q > 0$, be independent of X. Then $(\bar{G}_{\mathbf{e}_q}, S_{\mathbf{e}_q}) \perp\!\!\!\perp \left(\mathbf{e}_q - \bar{G}_{\mathbf{e}_q}, S_{\mathbf{e}_q} - X_{\mathbf{e}_q}\right)$. Furthermore, for $\alpha, \beta > 0$,*

$$\mathbb{E}\left[e^{-\alpha\bar{G}_{\mathbf{e}_q}-\beta S_{\mathbf{e}_q}}\right] = \exp\left(\int_0^\infty\int_0^\infty \left(e^{-\alpha t-\beta x} - 1\right)\mathbb{P}\left(X_t \in dx\right) e^{-qt}\frac{dt}{t}\right) \tag{8.9}$$

and

$$\mathbb{E}\left[e^{-\alpha\left(\mathbf{e}_q-\bar{G}_{\mathbf{e}_q}\right)-\beta\left(S_{\mathbf{e}_q}-X_{\mathbf{e}_q}\right)}\right] = \exp\left(\int_0^\infty\int_0^\infty \left(e^{-\alpha t-\beta x} - 1\right)\mathbb{P}\left(-X_t \in dx\right) e^{-qt}\frac{dt}{t}\right). \tag{8.10}$$

Proof. From the last claim of Lemma 8.10 we get that

$$\left(\mathbf{e}_q, X_{\mathbf{e}_q}\right) = (\bar{G}_{\mathbf{e}_q}, S_{\mathbf{e}_q}) + \left(\mathbf{e}_q - \bar{G}_{\mathbf{e}_q}, -S_{\mathbf{e}_q} + X_{\mathbf{e}_q}\right) \stackrel{d}{=} (\bar{G}_{\mathbf{e}_q}, S_{\mathbf{e}_q}) + \left(\underline{G}_{\mathbf{e}_q}, I_{\mathbf{e}_q}\right).$$

Since all pairs $\left(\mathbf{e}_q, X_{\mathbf{e}_q}\right)$ is infinitely divisible from Lemma 8.10 then if $\mu^{\pm}$ are the Lévy measures of the pairs involving the supremum then

$$t^{-1}e^{-qt}\mathbb{P}\left(X_t \in dx\right)\, dt = \mu^+(dt, dx) + \mu^-(dt, dx)$$

with $\mu^+(dt,(-\infty,0]) = \mu^-(dt,[0,\infty)) = 0$. Therefore, $\mu^\pm$ are the respective restriction of $t^{-1}e^{-qt}\mathbb{P}\left(X_t \in dx\right) \, dt$ on $(0,\infty), (-\infty,0)$ in the spatial coordinate. This gives (8.9) and (8.10). □

Combining (8.9) and (8.10), taken with $q=1$, with Lemma 8.10, we obtain

$$\begin{aligned} \frac{\kappa(1,0)}{\kappa(1+\alpha,\beta)} &= \exp\left(\int_0^\infty \int_0^\infty \left(e^{-(1+\alpha)t-\beta x} - e^{-t}\right) \mathbb{P}\left(X_t \in dx\right) \frac{dt}{t}\right), \\ \frac{\widehat{\kappa}(1,0)}{\widehat{\kappa}(1+\alpha,\beta)} &= \exp\left(\int_0^\infty \int_0^\infty \left(e^{-(1+\alpha)t-\beta x} - e^{-t}\right) \mathbb{P}\left(-X_t \in dx\right) \frac{dt}{t}\right). \end{aligned}$$

Substituting $z = 1+\alpha$ and $\zeta = \beta$, and using analyticity for $\Re(z) > 0$, $\Re(\zeta) > 0$, we obtain

$$\begin{aligned} \kappa(z,\zeta) &= k \exp\left(\int_0^\infty \int_0^\infty \left(e^{-t} - e^{-zt-\zeta x}\right) \mathbb{P}\left(X_t \in dx\right) \frac{dt}{t}\right), \\ \widehat{\kappa}(z,\zeta) &= \widehat{k} \exp\left(\int_0^\infty \int_0^\infty \left(e^{-t} - e^{-zt-\zeta x}\right) \mathbb{P}\left(-X_t \in dx\right) \frac{dt}{t}\right), \end{aligned} \tag{8.11}$$

where $k = \kappa(1,0)$ and $\widehat{k} = \widehat{\kappa}(1,0)$. These are known as Fristedt's formulae. Both local times can be normalised so that $k\widehat{k} = 1$. The expressions can be analytically or continuously extended whenever the integrals in the exponents admit the same extension. In particular, setting $\zeta = 0$ gives

$$\begin{aligned} \kappa(q,0) &= k \exp\left(\int_0^\infty \left(e^{-t} - e^{-qt}\right) \mathbb{P}\left(X_t > 0\right) \frac{dt}{t}\right), \\ \widehat{\kappa}(q,0) &= \widehat{k} \exp\left(\int_0^\infty \left(e^{-t} - e^{-qt}\right) \mathbb{P}\left(X_t < 0\right) \frac{dt}{t}\right). \end{aligned} \tag{8.12}$$

By Frullani's integral, and since $\mathbb{P}\left(X_t = 0\right) = 0$ for $t > 0$ when X is not a CPP, we obtain

$$\kappa(q,0)\widehat{\kappa}(q,0) = q. \tag{8.13}$$

Another application of Lemma 8.10 and the Wiener–Hopf factorization is obtained from

$$X_{\mathbf{e}_q} = S_{\mathbf{e}_q} + \left(X_{\mathbf{e}_q} - S_{\mathbf{e}_q}\right),$$

where the two summands are independent. Therefore,

$$\begin{aligned} \frac{q}{q - \Psi(i\theta)} &= \mathbb{E}\left[e^{i\theta X_{\mathbf{e}_q}}\right] \\ &= \frac{\kappa(q,0)}{\kappa(q,-i\theta)} \frac{\widehat{\kappa}(q,0)}{\widehat{\kappa}(q,i\theta)} \\ &= \frac{q}{\kappa(q,-i\theta)\widehat{\kappa}(q,i\theta)}, \end{aligned}$$

where in the last equality we used (8.13). Hence

$$\Psi(i\theta) - q = -\kappa(q,-i\theta)\widehat{\kappa}(q,i\theta). \tag{8.14}$$

This identity lies at the core of many modern developments in the theory of Lévy processes. It links the input quantity Ψ, which describes the Lévy process X, to the Lévy-Khintchine exponents of the ascending and descending ladder processes, which encode its fluctuations.

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